PHYSICS

1) If magnetic field passing through a coil of area 0.1 m^2 is changing according to the equation B = $10^{\sin\left(\frac{2\pi t}{3}\right)}$ tesla. Find the magnitude of induced emf at t = 0.5s

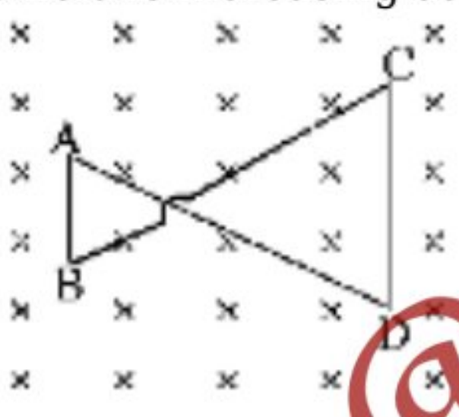
(1) $\frac{\pi}{2}$ volt

(2) $\frac{\pi}{3}$ volt

(3) $\frac{2\pi}{3}$ volt

(4) $\frac{\pi}{4}$ volt

2) A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are



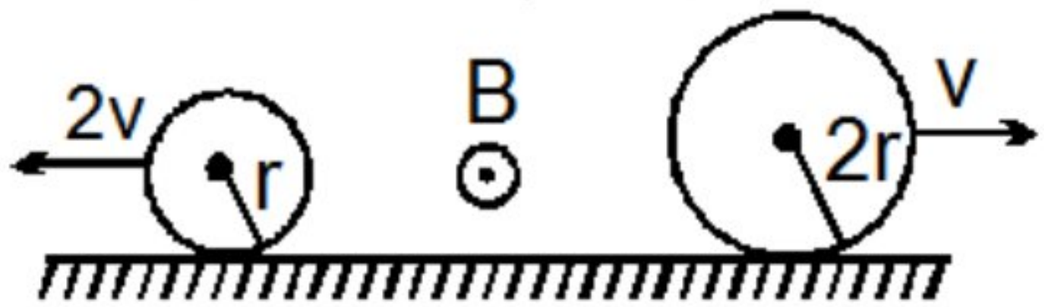
(1) B to A and D to C

(2) A to B and C to D

(3) A to B and D to C

(4) B to A and C to D

3) Two conducting rings P and Q of radii r and 2r rotate uniformly in opposite directions with centre of mass velocities 2v and v respectively on a conducting surface S. There is a uniform magnetic field of magnitude B perpendicular to the plane of the rings. The potential difference between the highest points of the two rings is



(1) Zero

(2) 4 Bvr

(3) 8 Bvr

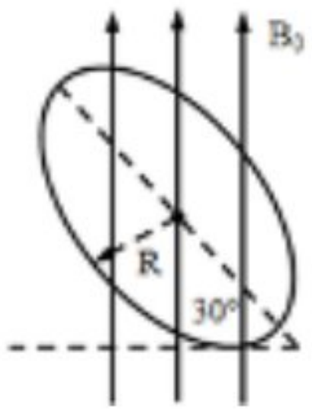
(4) 16 Bvr

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4) Circular surface of radius R makes an angle 30° with the horizontal and uniform magnetic field exists in vertically upward direction. Flux through the surface is –



(1) $\frac{\sqrt{3}}{2}\pi R^2 B_0$

(2) $\sqrt{3}\pi R^2 B_0$

(3) $\frac{1}{2}\pi R^2 B_0$

(4) $\pi R^2 B_0$

5) A long solenoid of cross–sectional radius a has a thin insulated wire ring tightly put on its winding. One half of the ring has a resistance n times that of other half. The magnetic induction produced by the solenoid varies with time as B = bt. Where b is a positive constant find magnitude of conservative electric field in ring.

(1) $\frac{n+1}{n-1}\cdot\frac{ab}{2}$

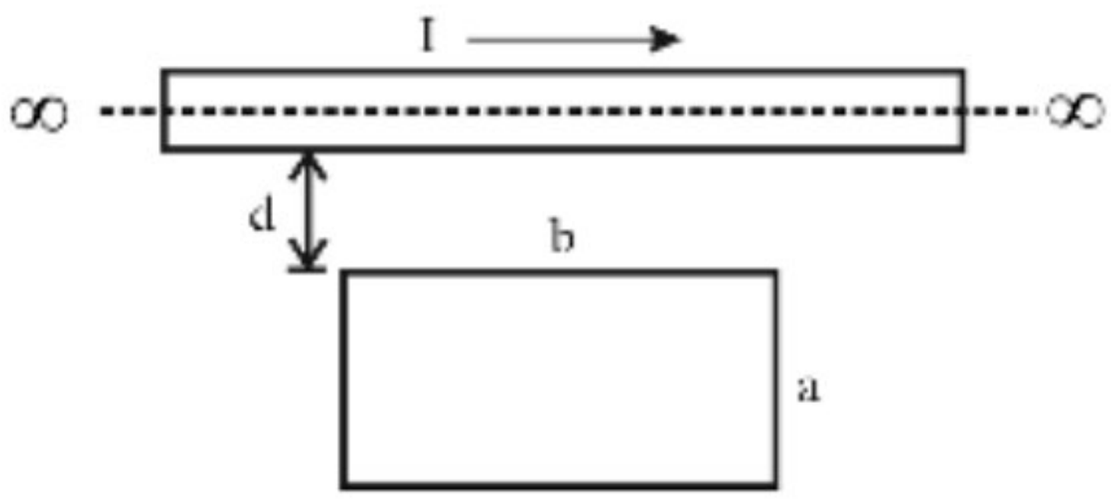
(2) $\frac{n-1}{n+1}\cdot\frac{ab}{2}$

(3) $\frac{ab}{2}$

(4) $n\cdot\frac{ab}{2}$

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6) A long straight wire is parallel to one edge as in figure. If the current in the long wire is varies in time as $I = I_0 e^{-t/\tau}$, what will be the induced emf in the loop ?



(1) $\frac{\mu_0 bI}{\pi\tau}\ell n\left(\frac{d+a}{d}\right)$

(2) $\frac{\mu_0 bI}{2\pi\tau}\ell n\left(\frac{d+a}{d}\right)$

(3) $\frac{2\mu_0 bI}{\pi\tau}\ell n\left(\frac{d+a}{d}\right)$

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(4) $\frac{\mu_0 bI}{\pi\tau}\ell n\left(\frac{d}{d+a}\right)$

7) A stretchable conducting ring is in the shape of a circle.It is kept in a uniform magnetic field (B) that is perpendicular to the plane of the ring. The ring is pulled out uniformly from all sides so as to increase its radius at a constant rate $\frac{dr}{dt} = V$ while maintaining its circular shape.The rate of work done by the external agent against the magnetic force when the radius of the ring is r, Resistance of the ring remains constant at R, is given as $\frac{4\pi^2 B^2 r^2 v^2}{xR}$ where x is_____

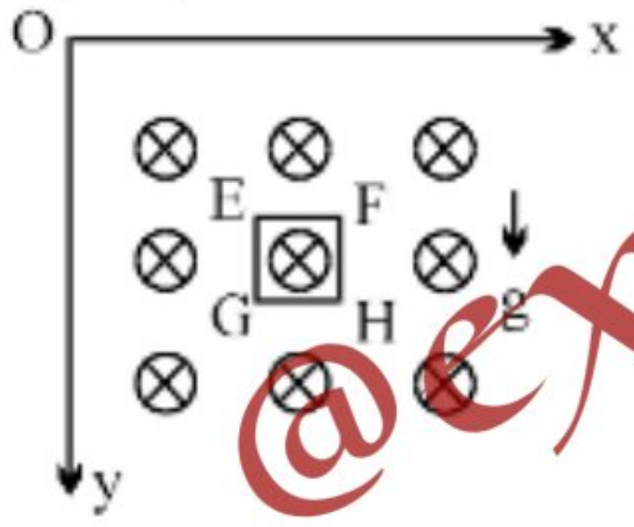
(1) 1

(2) 2

(3) 3

(4) 4

8) A magnetic field $B = (B_0 y / a)\hat{k}$ is into the plane of paper in the +z direction. B_0 and a are positive constants. A square loop EFGH of side a, mass m and resistance R, in x-y plane, starts falling under the influence of gravity. Note the directions of x and y axes in the figure. The induced current in the loop will be :-



(where v is the velocity at time t)

(1) $\frac{B_0 av}{R}$, In CW direction

(2) $\frac{B_0 av}{R}$, In ACW direction

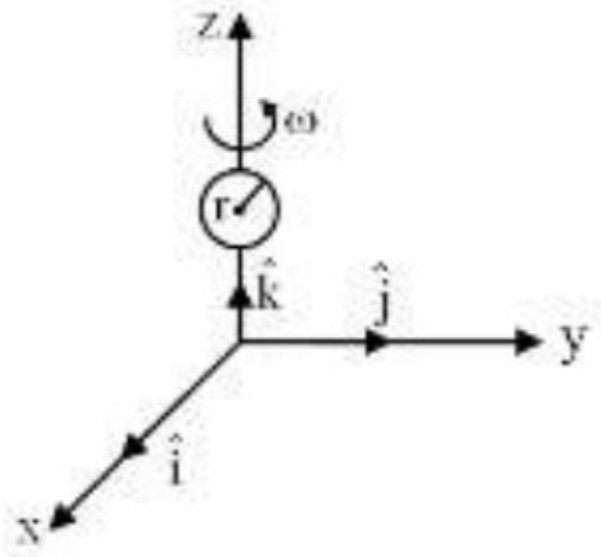
(3) $\frac{2B_0 av}{R}$, In ACW direction

(4) $\frac{B_0 av}{2R}$, In ACW direction

9) A circular loop wire of radius r rotates about the z–axis with angular velocity ω. The normal to the loop is always perpendicular to the z–axis. At time t = 0, the normal is parallel to the y–axis. An external consent magnetic field $\vec{B} = B_y\hat{j} + B_z\hat{k}$ is applied. The EMF ε(t) induced in the loop is :-

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(1) $\pi r^2 \omega B_y \sin \omega t$

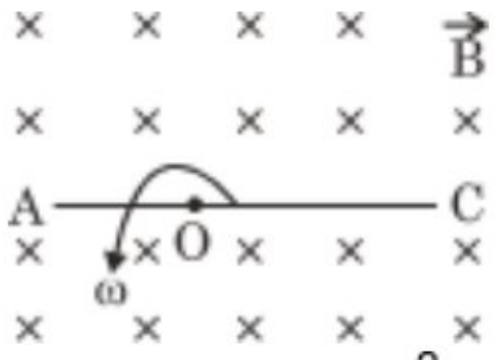
(2) $\pi r^2 \omega B_z \cos \omega t$

(3) $\pi r^2 \omega B_z \sin \omega t$

(4) $\pi r^2 \omega B_y \cos \omega t$

10)

A conducting rod AC of length 4□ is rotated about a point O in a uniform magnetic field $\vec{B}$ directed into the paper. AO = □ and OC = 3□. Then :-



(1) $V_A - V_0 = \frac{B\omega l^2}{2}$

(2) $V_0 - V_C = \frac{9}{2} B\omega l^2$

(3) $V_A - V_C = \frac{7}{2} B\omega l^2$

(4) $V_C - V_0 = \frac{9}{2} B\omega l^2$

11) A circular coil of 1000 turns each with area $1m^2$ is rotated about its vertical diameter at the rate of one revolution per second in a uniform horizontal magnetic field of 0.07T. The maximum voltage generation will be _____V.

(1) 440

(2) 220

(3) 110

(4) 460

12) A boat is moving due east in a region where the earth's magnetic field is 5.0×10^{-5} $NA^{-1}m^{-1}$ due north and horizontal. The boat carries a vertical aerial 3m long. If the speed of the boat is 3m/s, find the magnitude of the induced emf in the wire of aerial (in mV).

(1) 0.45

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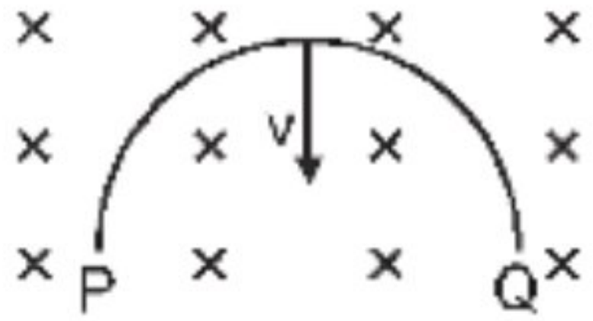
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(2) 0.63

(3) 0.71

(4) 0.92

13) A semicircle loop PQ of radius 'R' is moved with velocity 'v' in transverse magnetic field as shown in figure. The value of induced emf. at the end of loop is :–



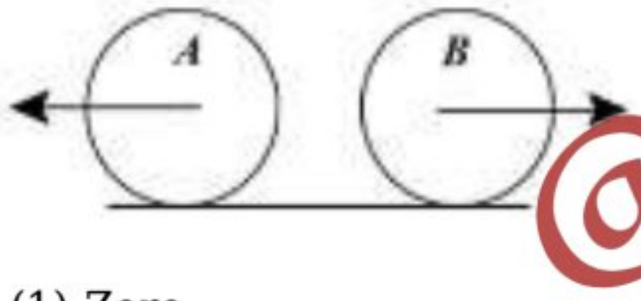
(1) Bv (πr), end 'P' at high potential

(2) 2 BRv, end P at high potential

(3) 2 BRv, end Q at high potential

(4) $B\frac{\pi R^2}{2}$ v, end P at high potential

14) Two identical conducting rings A & B of radius R are in pure rolling over a horizontal conducting plane with same speed (of center of mass)v but in opposite direction. A constant magnetic field B is present pointing inside the plane of paper. then the potential difference between the highest point of the two rings, is:



(1) Zero

(2) 2 Bvr

(3) 4 Bvr

(4) None of these.

15) A conducting metal circular-wire-loop of radius r is placed perpendicular to a magnetic field which varies with time as $B = B_0 e^{-t/\tau}$, where B_0 and τ are constants, at time t = 0. If the resistance of the loop is R then the heat generated in the loop after a long time ($t \to \infty$) is :

(1) $\frac{\pi^2 r^4 B_0^4}{2\tau R}$

(2) $\frac{\pi^2 r^4 B_0^2}{2\tau R}$

(3) $\frac{\pi^2 r^4 B_0^2 R}{\tau}$

(4) $\frac{\pi^2 r^4 B_0^2}{R\tau}$

16)

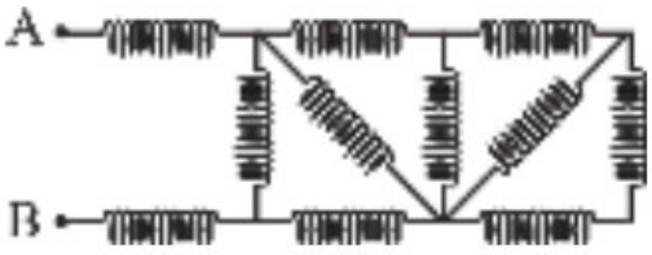
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A network of inductances, each of value 1 H, is shown in figure. The equivalent inductance of the circuit between points A and B is :-



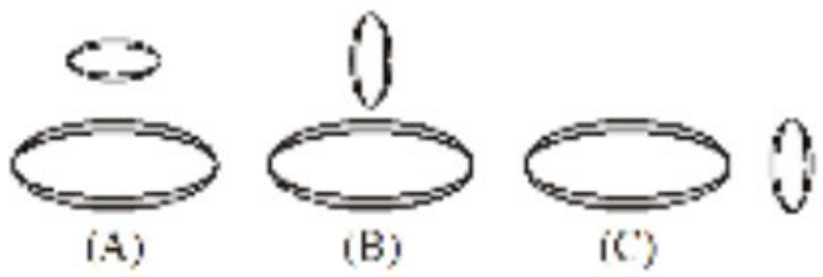
(1) 6.218 H

(2) 0.268 H

(3) 8.162 H

(4) 2.618 H

17) Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be :-



(1) Maximum in situation (A)

(2) Maximum in situation (B)

(3) Maximum in situation (C)

(4) The same in all situations

18) A time varying voltage $V = 3t^2$ is applied across an ideal inductor of L = 2H. The potential energy at time t=1 sec in magnetic field is increasing at rate of X J/s. Calculate value of X. (Initial current in inductor is zero)

(1) 1.5

(2) 3

(3) 2.5

(4) 4

19) A non-uniform solenoid has a length L and a variable turn density $n(x) = n_0\frac{x}{L}$ where x is the distance from one end $(0 \leq x \leq L)$. Assuming the radius R is very small (R << L), the selfinductance of this setup is proportional to:

(1) $n_0^2 L$

(2) $\frac{1}{3}n_0^2 L$

(3) $\frac{1}{2}n_0^2 L$

(4) $\frac{1}{4}n_0^2 L$

20) A thin wire of length l is wound into a long single-layer solenoid of radius R. If the wire is rewound to form a solenoid of radius R/2 keeping the tight-packing condition intact (length adapts accordingly), the self-inductance will change by a factor of:

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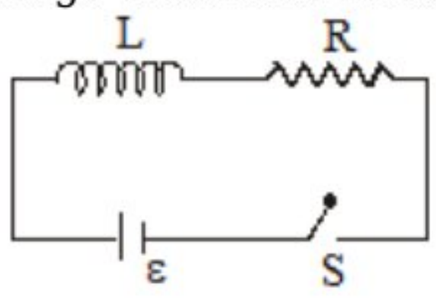
(1) 2

(2) 4

(3) $\frac{1}{2}$

(4) 1

21) A shown in the figure, a battery of emf e is connected to an inductor L and resistance R in series. The switch is closed at t = 0. The total charge that flows from the battery, between t = 0 and $t = t_c$ (t_c is the time constant of the circuit) is :



(1) $\frac{\varepsilon L}{R^2}\left(1-\frac{1}{e}\right)$

(2) $\frac{\varepsilon R}{eL^2}$

(3) $\frac{\varepsilon L}{R^2}$

(4) $\frac{\varepsilon L}{eR^2}$

22) The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjusting the separation between windings. The inductance of solenoid will be proportional to :

(1) $1/L^2$

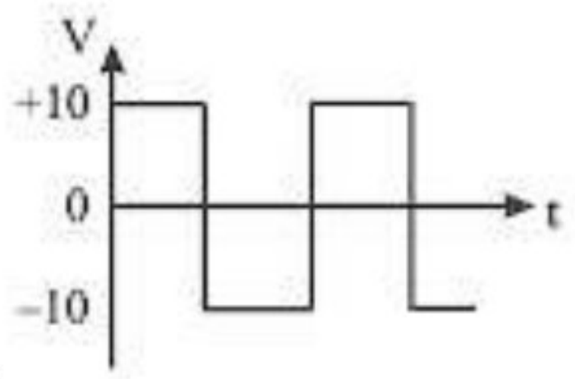
(2) $1/L$

(3) L

(4) L^2

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23) The r.m.s. voltage of the wave form shown is :



(1) 10 V

(2) 7 V

(3) 6.37 V

(4) None of these

24) Which one of the following ratios is always less than unity in a transformer ?

(1) Voltage transformation ratio

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(2) Current transformation ratio

(3) Power transformation ratio

(4) Ratio of turns in the windings.

25) A loss free transformer has 500 turns on its primary winding and 2500 in secondary. The metres of the secondary indicate 200 volts at 8 amperes under these conditions. The voltage and current in the primary is

(1) 100 V, 16 A

(2) 40 V, 40 A

(3) 160 V, 10 A

(4) 80 V, 20 A

26) A 750 Hz, 20 V (rms) source is connected to a resistance of 100 Ω, an inductance of 0.1803 H and a capacitance of 10 μF all in series. The time in which the resistance (heat capacity 2J/°C) will get heated by 10°C. (assume no loss of heat to the surroundings) is close to :

(1) 418 s

(2) 245 s

(3) 348 s

(4) 365 s

27) Given below are two statements :

Statement-I **:** The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement-II : In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below :

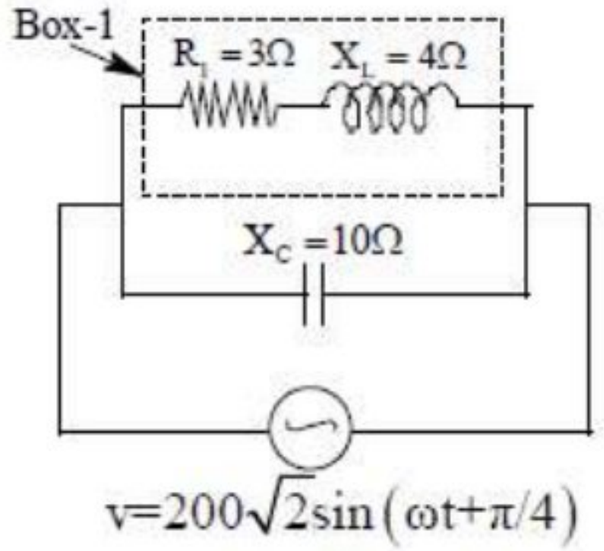
(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II in false.

(4) Statement I is false but Statement II is true.

28) As shown in figure , Instantaneous current in branch having capacitor C will be : (given $\tan^{-1}$



(4/3) = 53°)

(1) $20\sqrt{2}\sin\left(\omega t + \frac{3\pi}{4}\right)$

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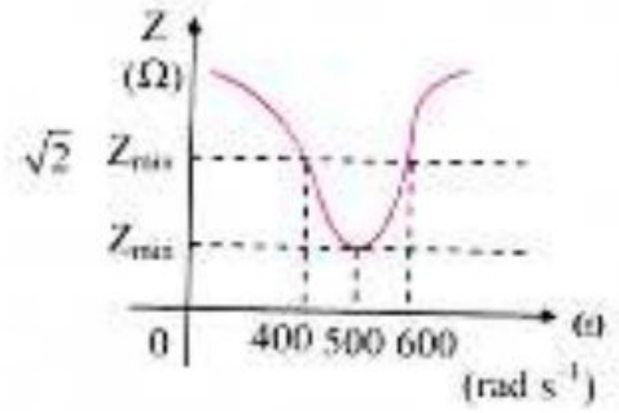
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(2) $40\sqrt{2}\sin\left(\omega t+\frac{\pi}{4}\right)$

(3) $60\sqrt{2}\sin\left(\omega t-\frac{\pi}{4}\right)$

(4) None of these

29) Total impedance (z) of a series LCR circuit varies with angular frequency (ω) of the a.c. source connected to it as shown in the graph. The quality factor (Q) of the series LCR circuit is



(1) 2.5

(2) 0.4

(3) 5

(4) 1

30) Which is a choke coil (connected to ac source)

(1) L-C series circuit

(2) L-R series circuit (high L and low R)

(3) LCR series circuit (low L)

(4) L-R series circuit (high R)

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31) A complex current wave is given by
i = 5 + 5 sin (100 ωt) A. Its average value over one time period is given as :-

(1) 10 A

(2) 5 A

(3) $\sqrt{50}$ A

(4) 0

32)

An alternating current is given by the equation $i = i_{01}\cos\omega t + i_{02}\sin\omega t$. The r.m.s. current is given by :-

(1) $\frac{1}{\sqrt{2}}(i_{01}+i_{02})$

(2) $\frac{1}{\sqrt{2}}(i_{01}+i_{02})^2$

(3) $\frac{1}{\sqrt{2}}(i_1^2+i_2^2)^{1/2}$

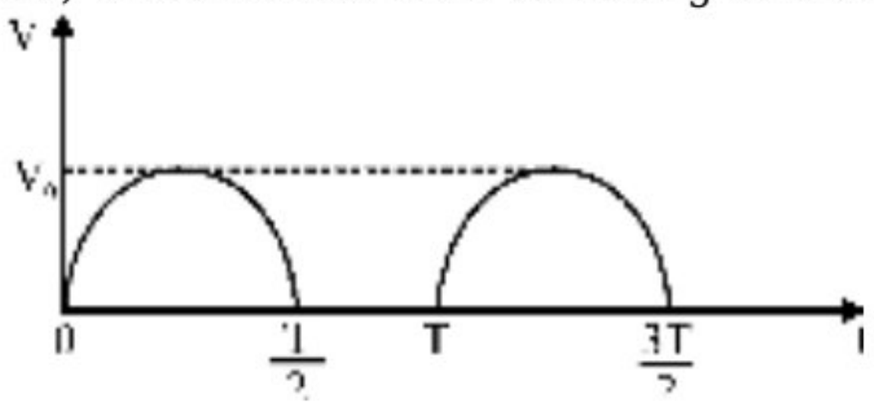
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(4) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$

33) What will be r.m.s. value of given. Voltage over one cycle.



(1) V_0

(2) $\frac{V_0}{\sqrt{2}}$

(3) $\frac{V_0}{2}$

(4) $\frac{V_0}{4}$

34) A capacitor of capacitance 100 μF & a resistance of 100Ω is connected in series with AC supply of 220V, 50 Hz. The current leads the voltage by ..... :-

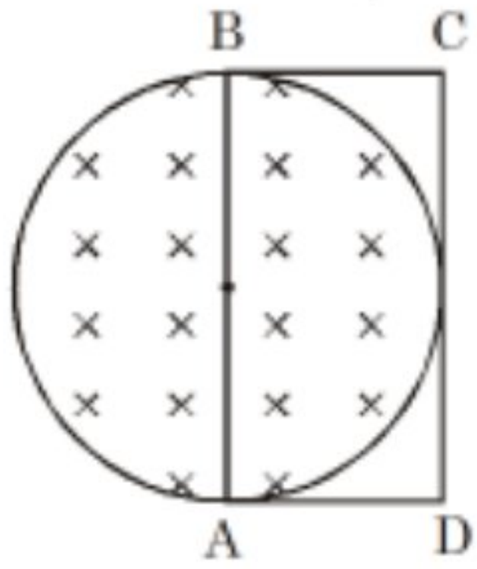
(1) $\tan^{-1}\left(\frac{1}{2\pi}\right)$

(2) $\tan^{-1}\left(\frac{1}{\pi}\right)$

(3) $\tan^{-1}\left(\frac{2}{\pi}\right)$

(4) $\tan^{-1}\left(\frac{4}{\pi}\right)$

35) There exists a uniform but time varying magnetic field B = a + bt normal to plane of paper in a cylindrical region of radius 'R' as shown. A rectangular conducting loop is placed as shown. Induced



emf in branches AB and BC are :

(1) $\pi R^2 b/2, 0$

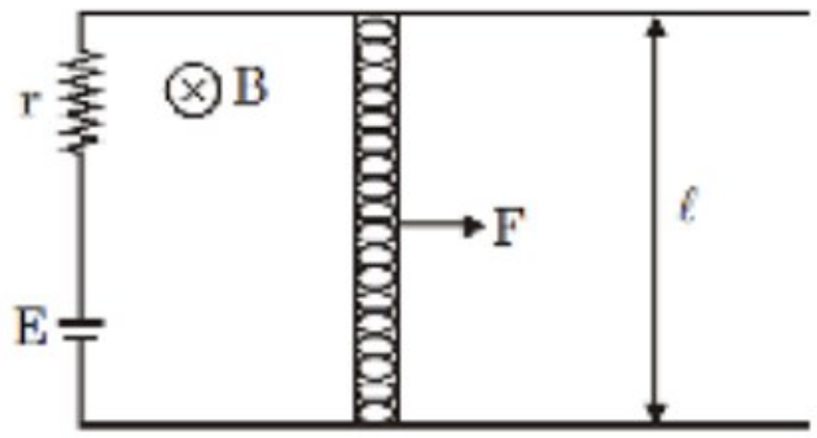
(2) 2R2b, 0

(3) $0, \pi R^2 b/8$

(4) $0, \pi R^2 b/4$

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36) Two parallel conducting rails are connected to a source of emf E and internal resistance r. Another conducting rod of length ℓ having negligible resistance lies at rest and can slide without friction over the rails. A uniform magnetic field B is applied perpendicular to the plane of the rails. At t = 0, the rod is pulled along the rails by applying a force F. The velocity of the rod is observed to be $v = v_0 \cos(\omega t)$ then find the average power (in watt) spent by the force over 1 cycle. (Given B = 2 Tesla, $r = 2 \times 10^{-4}\Omega$, $v_0 = 2ms^{-1}$, $\ell = 1$ cm)



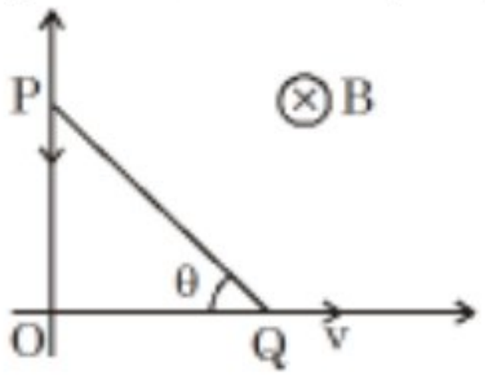
(1) 4

(2) 2

(3) 3

(4) 7

37) A conducting rod PQ of length ℓ moves between a fixed L-shaped conducting frame placed horizontally. A vertical magnetic field is also present as shown in figure. Point Q of rod is pulled horizontally with speed v while initially θ being 90°. At some instant t_0, current in loop becomes zero. (ℓ = 2m, v = 1 m/s) Angular velocity of rod at that instant is ω_0. Fill value of $\omega_0 \times t_0$.



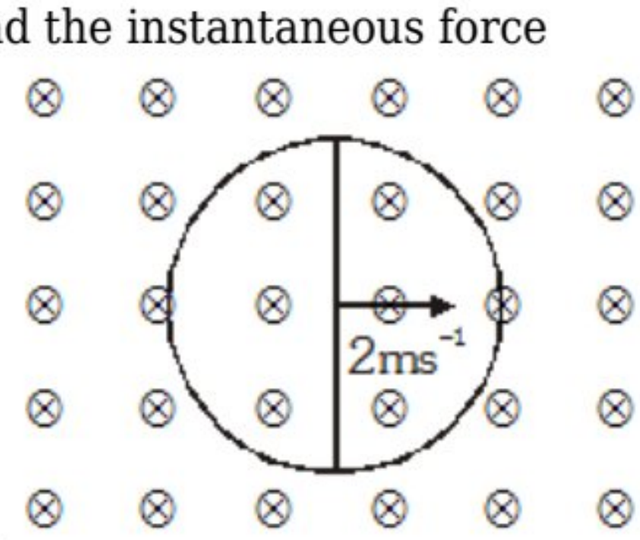
(1) 1

(2) 2

(3) 4

(4) 7

38) A circular loop of radius 2 m is kept in a magnetic field of strength 2T (plane of loop is perpendicular to the direction of magnetic field). Resistance of the loop wire is $\frac{2}{\pi}\Omega m^{-1}$. A conductor of length 4m is sliding with a speed 2 ms^{-1} as shown in the figure. Find the instantaneous force acting on the rod : (Assuming the rod to have a negligible resistance)



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(1) 8 N

(2) 16 N

(3) 32 N

(4) 64 N

39) A person is riding a cycle at a speed of 18 km/hr along the equator. The wheel consists of 18 spokes and has a diameter of 50 cm. A bulb is attached between centre of wheel and its periphery. If the resistance of bulb is 0.3 μΩ, then power dissipated in the bulb will be (given $\vec{B}_{earth} = 25\mu T$ & assume no slipping) :-

(1) 14.65 mW

(2) 1.628 mW

(3) 0.814 mW

(4) 0 mW

40) A fixed long ideal solenoid is encircled by a coaxial circular loop of radius R greater than radius of the solenoid. A current I is passed through the circular loop. In which of the following cases, emf is induced in the solenoid ?



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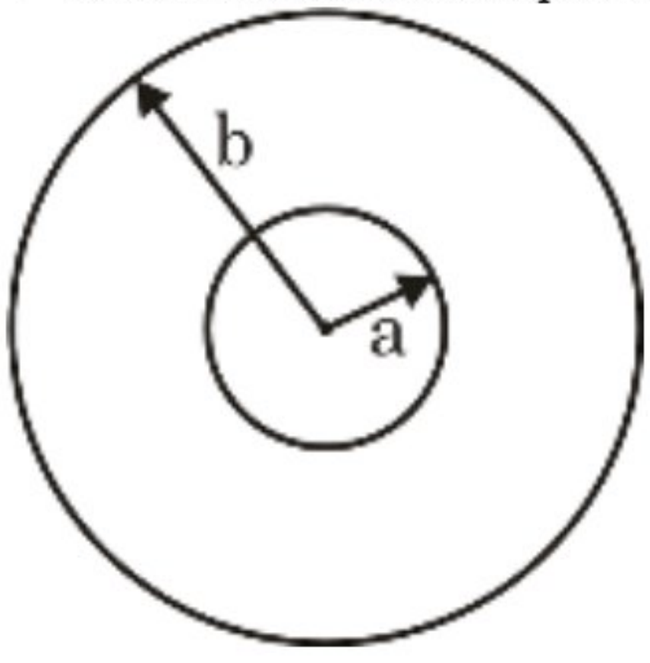
(1) The current in the loop is held constant and the loop is stationary.

(2) The current in the loop is $I = I_0 \sin\omega t$ and the loop is stationary.

(3) The current in the loop is constant and the loop performs angular SHM about its diameter (y axis) (but does not reach the solenoid surface)

(4) The current in the loop is constant and the loop performs SHM about its mean position in a direction perpendicular to the axis (but does not cross the solenoid)

41) The current in the outer coil is varying with time as $I = 2t^2$. If the resistance of the inner coil is R and b >> a then the heat developed in the inner coil between t = 0 and t seconds is $\frac{4\mu_0^2\pi^2a^4t^3}{kb^2R}$.



Find k.

(1) 3

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(2) 4

(3) 6

(4) 8

42) For a sinusoidal AC current $i = I_m \sin(\omega t)$, the value of current averaged over one complete cycle is:

(1) $2I_m/\pi$

(2) $I_m/\sqrt{2}$

(3) 0

(4) I_m

43) The heat produced in a resistor by an AC source of peak voltage V_m is H. If the peak voltage is doubled and frequency is halved, the new heat H′ produced in the same time will be:

(1) 2H

(2) 4H

(3) H/2

(4) H

44) Which of the following graphs correctly represents the variation of instantaneous current i with time t for an AC source?

(1) A linear graph passing through the origin.

(2) A parabolic graph.

(3) A sinusoidal graph.

(4) An exponential decay graph.

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45) Why are AC voltages preferred over DC for long-distance power transmission through resistors (wires)?

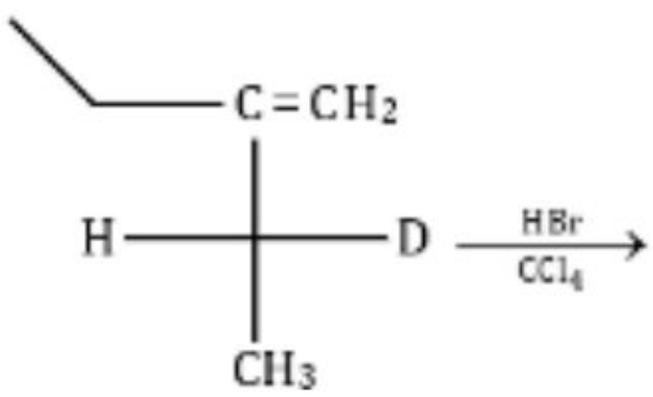
(1) AC resistance is lower than DC resistance

(2) AC can be stepped up to high voltages to minimize I^2R losses.

(3) AC current flows only on the surface of the wire

(4) Resistors don't heat up with AC

CHEMISTRY



1) Product (without rearrangement of carbocation)

What is the stereochemistry of product formed :

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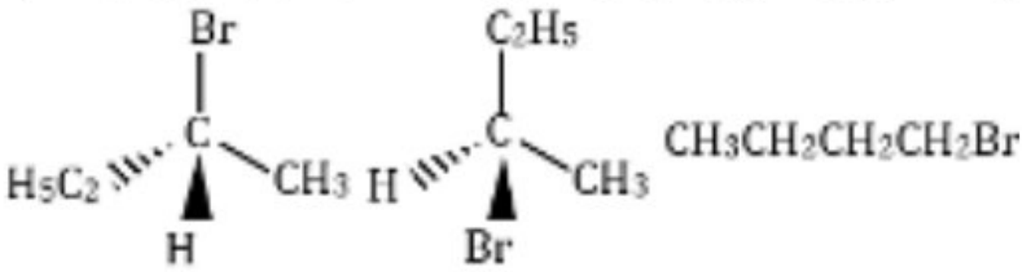
(1) Racemic mixture

(2) Optically inactive

(3) Mixture of diastereomers

(4) Meso product

2) The addition of HBr to 1-butene gives a mixture of products A, B and C



The mixture consists of

(1) A and B as major and C as minor products

(2) B as major, A and C as minor products

(3) B as minor, A and C as major products

(4) A and B as minor and C as major products

3) What is the best way to carry out the following transformation?
1-pentyne —→ pentanal

(1) $HgSO_4/H_2SO_4$

(2) H_2/Lindlar's catalyst; O_3; Zn-H_2O

(3) HIO_4/H_2O

(4) BH_3; H_2O_2/OH^-

4) In which of the following reaction, Markovnikov rule type is violated?

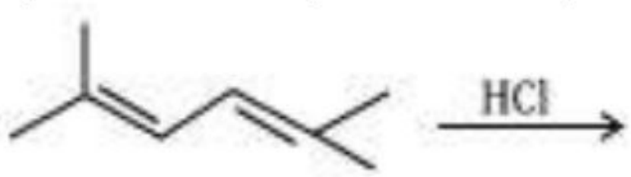
(1) $CH_3—O—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

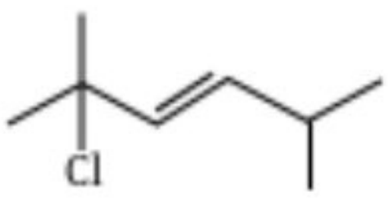
(2) $CH_3—NH—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

(3) $CH_3—S—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

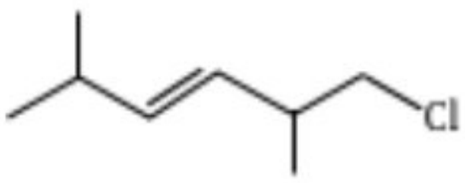
(4) $O_2N—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

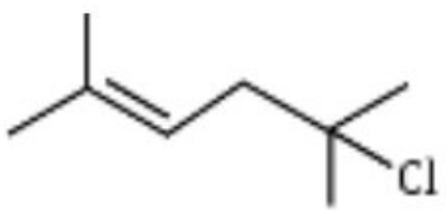
5) What is the product of 1, 4-addition in the reaction shown below?

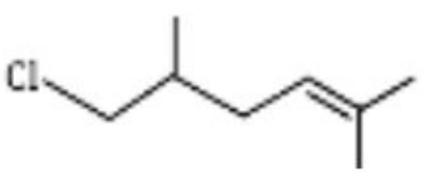


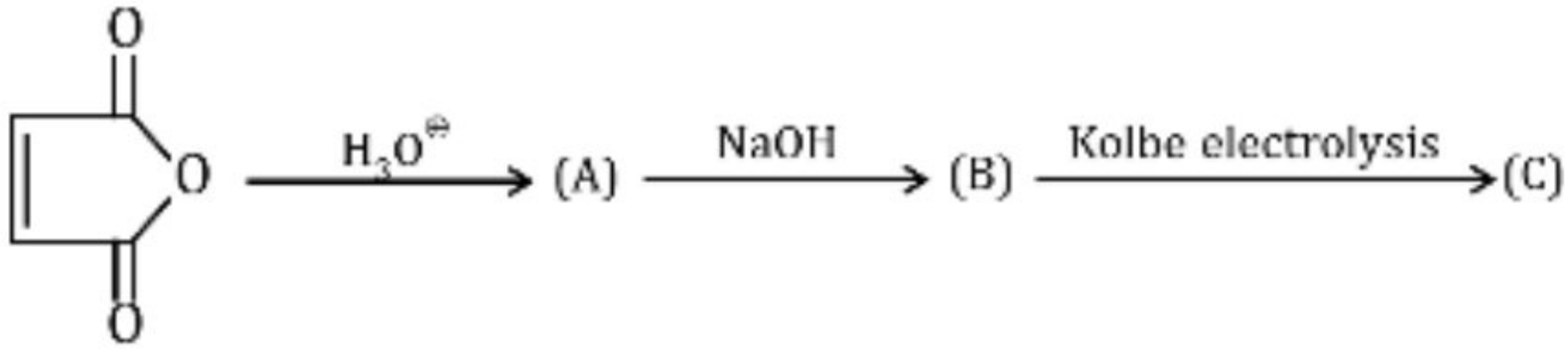
(1) 

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(2) 

(3) 

(4) 

6) 

Product (C) of above reaction is:

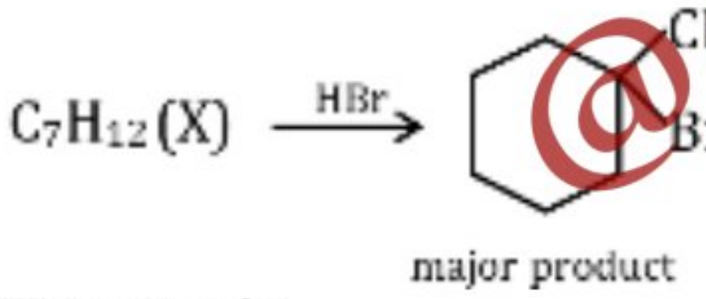
(1) $H_2C = CH_2$

(2) $CH_3 — C \equiv C — CH_3$

(3) $HC \equiv CH$

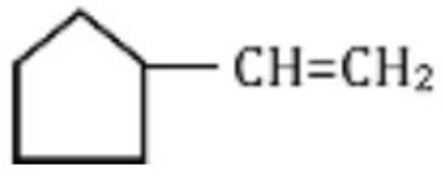
(4) $CH_3 — CH = CH — CH_3$

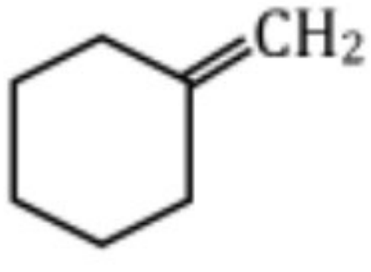
7) In the given reaction

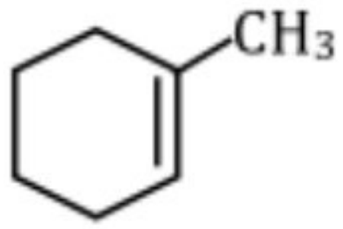


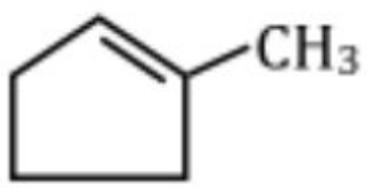
major product

(X) can not be :

(1) 

(2) 

(3) 

(4) 

8) Propene is allowed to react with B_2D_6 and the product is treated with acetic acid. The final product obtained is-

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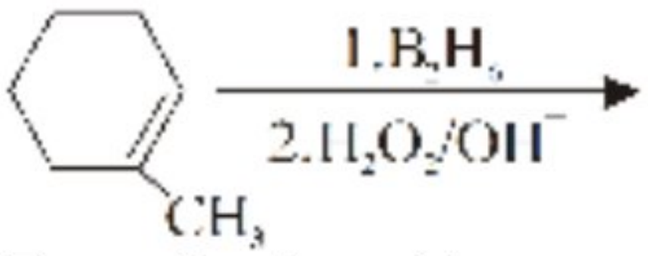
(1) 1–deuteriopropane

(2) 2–deuteriopropane

(3) 1–deuteriopropene

(4) 2–deuteriopropene

9) 1–Methylcyclohexene is allowed to react with B_2H_6. The product is then treated with H_2O_2 and NaOH. The reaction is -



The product formed is

(1) 1–methylcyclohexanol

(2) 2–methylcyclohexanol

(3) methylcyclohexane

(4) cyclohexanol

10) In the addition of HBr to propene in the absence of a peroxide, the first step involves the addition of -

(1) H^+

(2) Br^-

(3) $H^\bullet$

(4) $Br^\bullet$

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11) The addition of HCl to 1–phenylpropene gives-

(1) $C_6H_5CHClCH_2CH_3$

(2) $C_6H_5CH_2CHClCH_3$

(3) $C_6H_5CH_2CH_2CH_2Cl$

(4) $C_6H_5CH(CH_3)CH_2Cl$

12) The ease of formation of free radicals follows the order -

(1) $3° > 2° > 1° > \dot{C}H_3$

(2) $\dot{C}H_3 > 1° > 2° > 3°$

(3) $1° > 2° > 3° > \dot{C}H_3$

(4) $2° > 1° > 3° > \dot{C}H_3$

13) The Wurtz-Fittig reaction involves condensation of :

(1) two molecules of aralkyl-halides

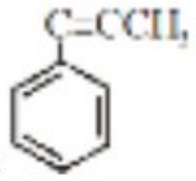
(2) one molecule of each aryl-halide and phenol

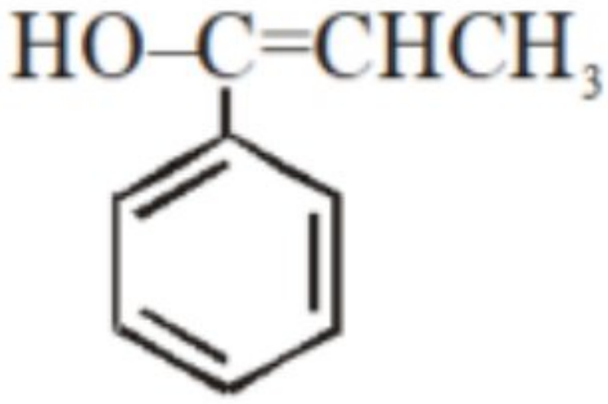
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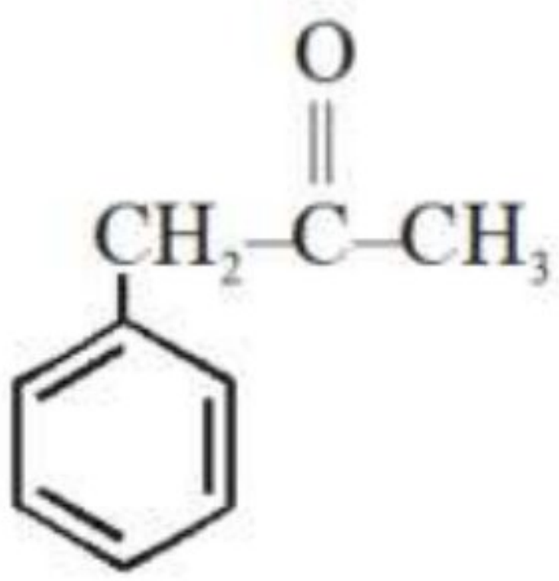
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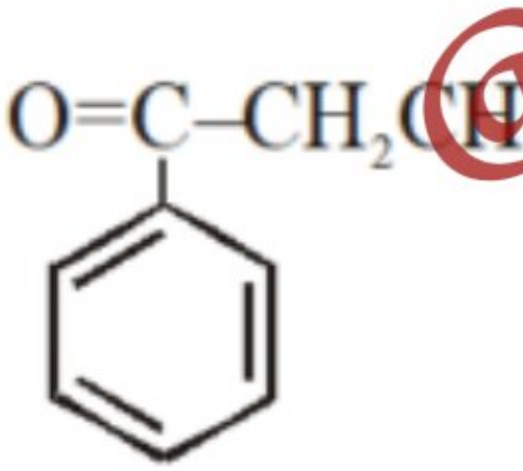
(3) two molecules of aryl halides

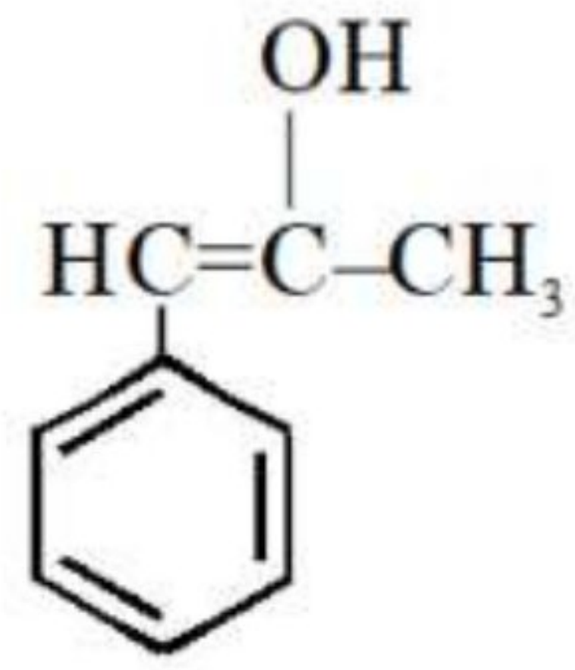
(4) one molecule of each of aryl-halide and alkyl-halide

14) In the given reaction  $\xrightarrow{H^+/Hg^{2+}}$ A the product 'A' is :-

(1) 

(2) 

(3) 

(4) 

15) The hydration of propyne results in formation of :-

(1) Propanol–1

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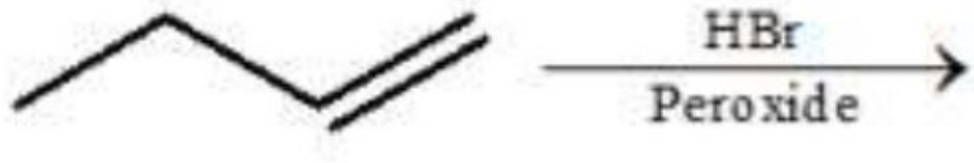
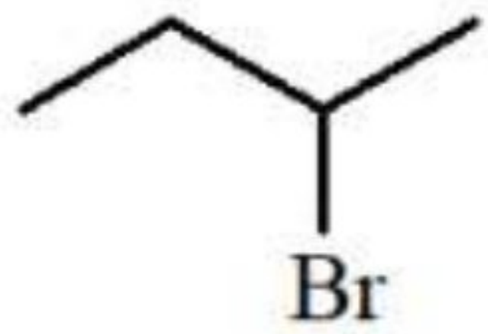
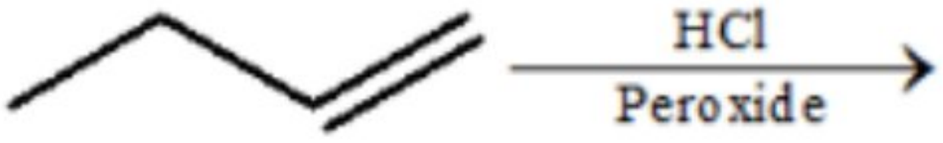
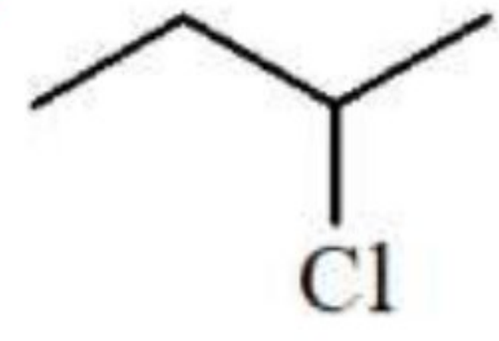
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(2) Propanal

(3) Acetone

(4) Propene

16) Match the Column.

	Column - I		Column - II
(A)		(p)	
(B)		(q)	
(C)		(R)	
(D)		(s)	

(1) A → Q;B → P;C → S;D → S

(2) A → P;B → Q;C → S;D → S

(3) A → Q;B → P;C → R;D → S

(4) A → Q;B → P;C → S;D → R

17) Which of the following reactions will result in the formation of a chiral centre in the product -

(1) $CH_3CH=CHCH_3+HBr\rightarrow$

(2) $CH_3CH=CH_2+HOBr\rightarrow$

(3) $CH_3CH_2CH=CH_2+HBr \xrightarrow{H_2O_2}$

(4) $CH_3CH_2CH=CH_2+HBr\rightarrow$

18) Which reagent converts propene to 1–propanol

(1) H_2O, H_2SO_4

(2) B_2H_6, H_2O_2, OH^-

(3) $Hg(OAc)_2$,$NaBH_4/H_2O$

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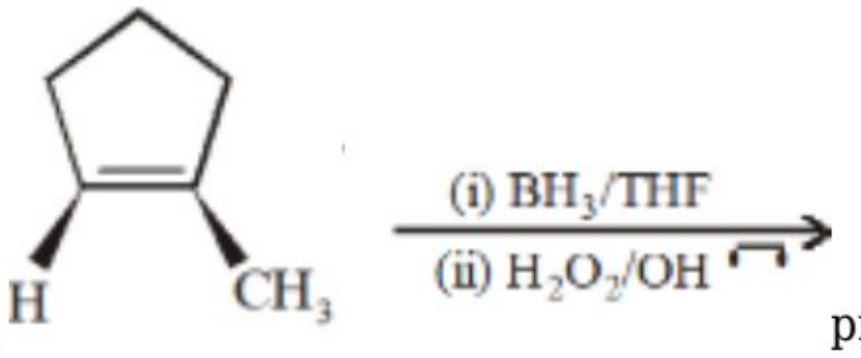
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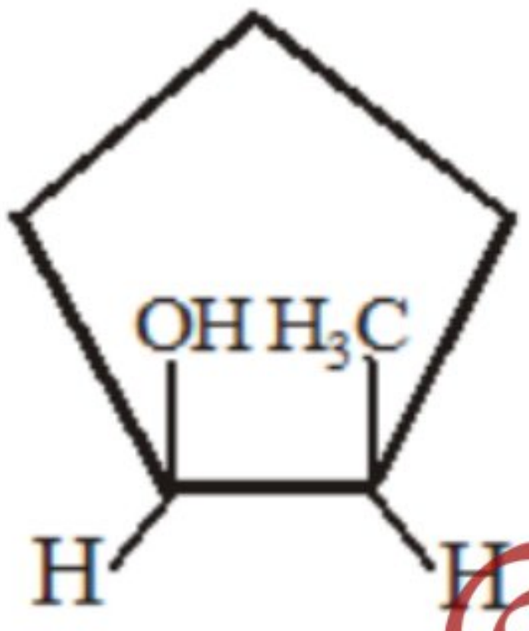
(4) Aq. KOH

19) cis–2–Butene on reaction with Br_2 in CCl_4 produces mainly -

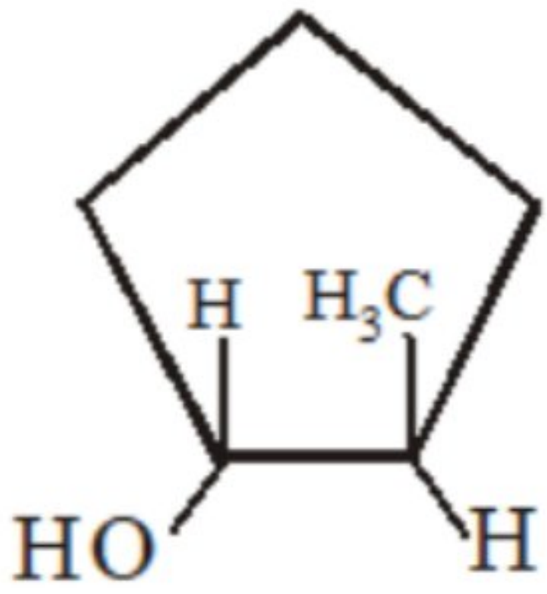
(1) 1–bromo–2–butene
(2) 2,2–dibromobutane
(3) meso–2,3–dibromobutane
(4) (±) 2,3–dibromobutane

20) What will be the product of following reaction 

product :

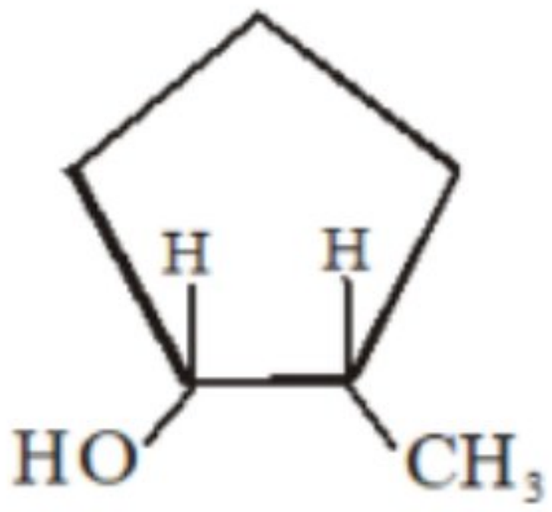
(1) 

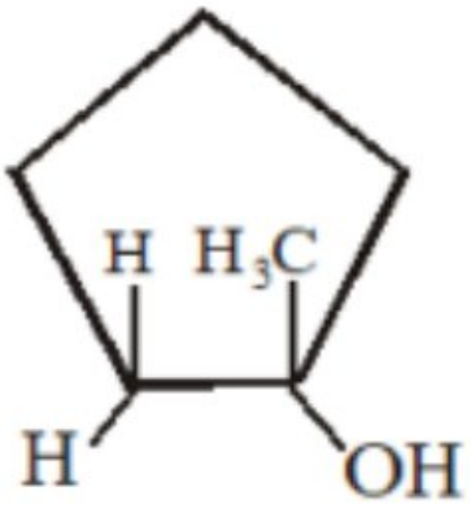
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(2) 

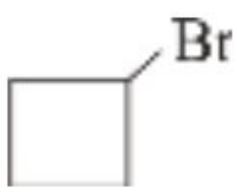
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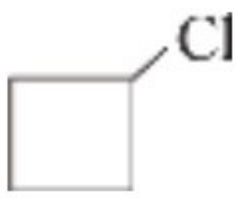
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(3) 

(4) 

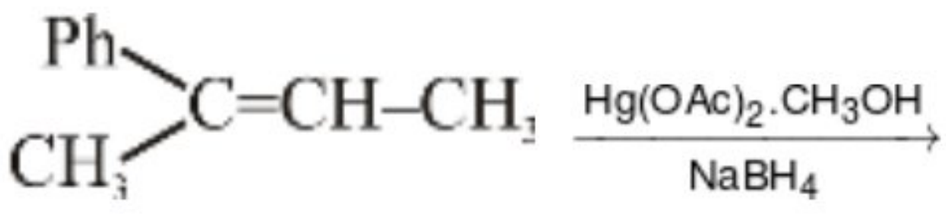
21) 1–Bromo–3–chloro cyclobutane on reaction with 2–equivalent of sodium in ether gives -

(1) 

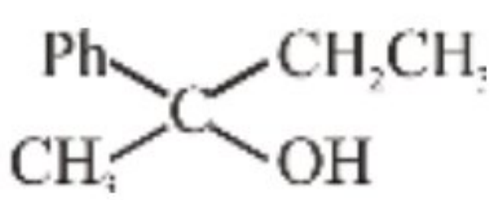
(2) 

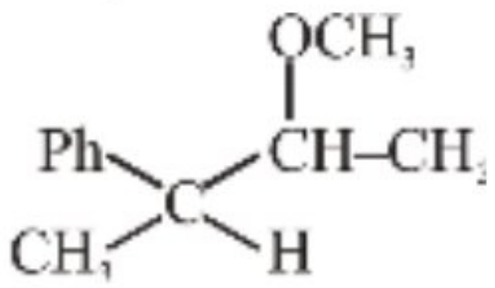
(3) 

(4) 

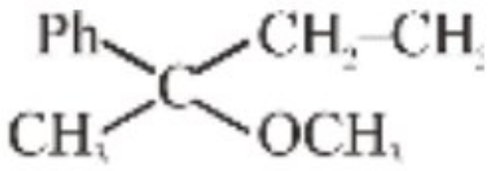
22) 

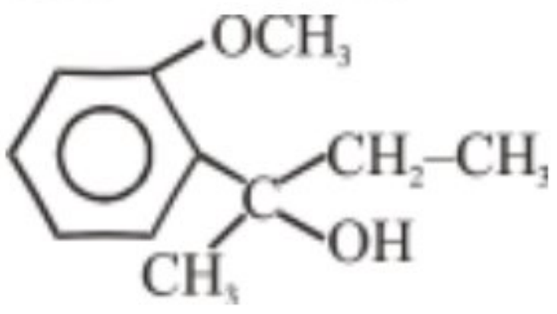
Product is :-

(1) 

(2) 

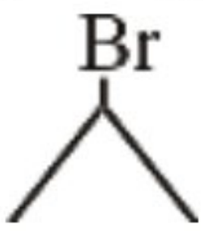
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(3) 

(4) 

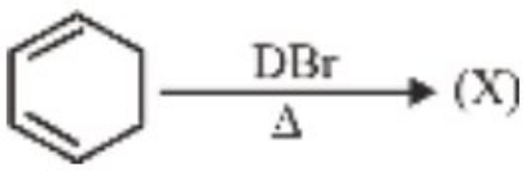
23) Which of the following alkyl bromides may be used for the synthesis of 2,3–dimethyl butane by Wurtz reaction ?

(1) 

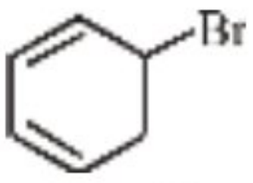
(2) 

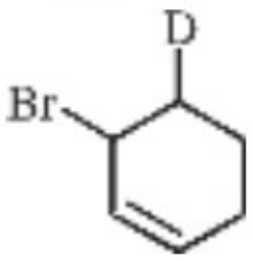
(3) Isobutyl bromide

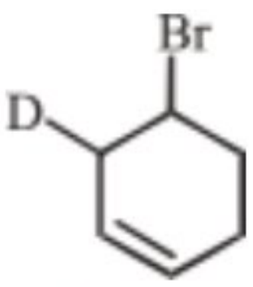
(4) s–butyl bromide

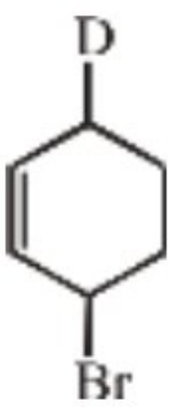
24) 

X is :-

(1) 

(2) 

(3) 

(4) 

25) The reaction $CH{\equiv}CH \longrightarrow CHCl_2CHO$ takes place in

(1) $CH{\equiv}CH \xrightarrow{HOCl}$

(2) $HC{\equiv}CH \xrightarrow{HCl\,/\,peroxide}$

(3) $CH{\equiv}CH \xrightarrow{Cl_2}$

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(4) $CH \equiv CH \xrightarrow{Cl_2 / \text{oxidizing agent}}$

26) When 2-Methylbut-2-ene is treated with hydrogen chloride, the major product obtained is

(1) 2-Chlorobutane
(2) 2-Chloro-2-methylbutane
(3) 3-Chloro-2-methylbutane
(4) 2-Chloro-3-methylbutane

27) In the reaction,

$$2nR - X \xrightarrow[\text{Dry ether}]{+2nNa} \text{product}$$

The product obtained is

(1) 2n-alkene
(2) n-sodium halide
(3) n-alcohol
(4) n-alkane

28) What is the number of moles of sp^2 hybrid carbon atoms in one mole of hexa-1,4-diyne?

(1) 5
(2) 3
(3) 4
(4) Zero

29) Which of the following is NOT obtained when a mixture of bromomethane and bromoethane is treated with sodium in dry ether?

(1) Propane
(2) Butane
(3) methane
(4) Ethane

30) How many gram of sodium (atomic mass 23u) is required to prepare one mole of ethane from methyl chloride by Wurtz reaction?

(1) 2
(2) 23
(3) 11.5
(4) 46

31) What is the molar mass of product hydrocarbon when 2 moles of methyl bromide reacts with large excess of sodium in dry ether?

(1) 15.0 g mol^{-1}
(2) 30.0 g mol^{-1}

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NOVA

(3) 7.5 g mol^{-1}

(4) 40.0 g mol^{-1}

32) Identify the correct stability order of following alkenes.

(i) $(CH_3)_2C = C(CH_2)_2$

(ii) $(CH_3)_2C = CH_2$

(iv) $(CH_3)_2C = CHCH_3$

(1) iii > ii > i

(2) i > iii > ii

(3) ii > i > iii

(4) i > ii > iii

33) By Wurtz reaction, a mixture of methyliodide and ethyliodide gives :

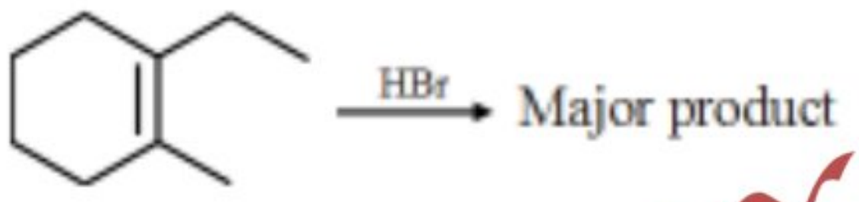
(1) Propane

(2) Ethane

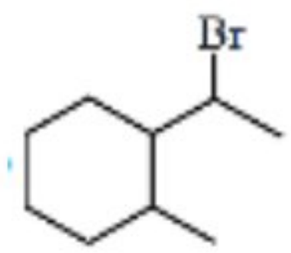
(3) Butane

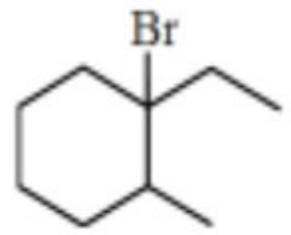
(4) A mixture of the all

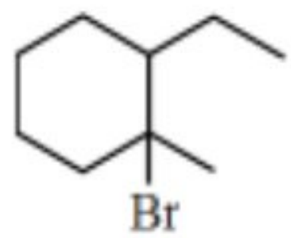
34)

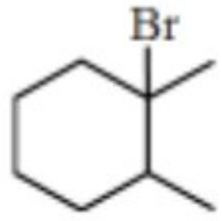


Major product is :-

(1)


(2)


(3)


(4)


35) During electrolysis which anion migrates towards anode so as to produce 2,3-dimethyl butane -

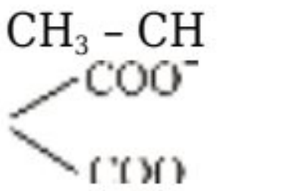
(1) CH_3-CH_2–COO^-

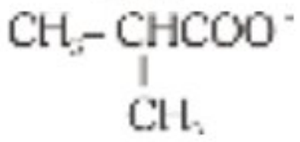
@examlix

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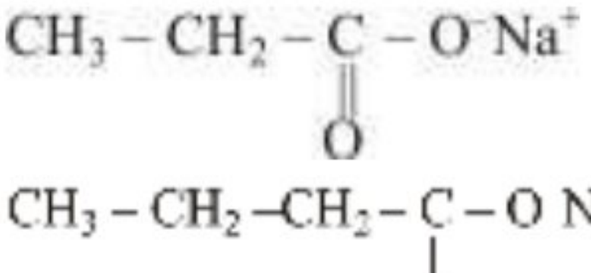
NOVA

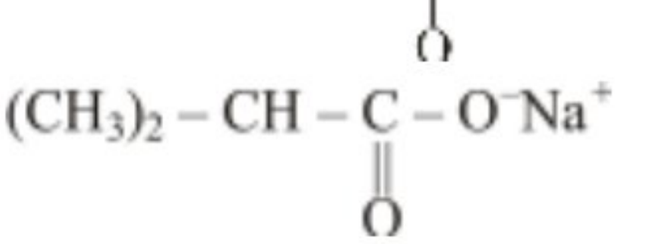
(2) 

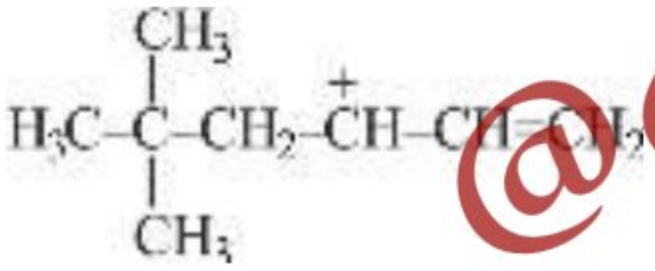
(3) 

(4) CH_3COO^-

36) Which sodium salt will be heated with sodalime to obtain propane -

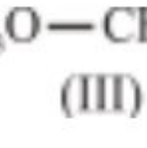
(1) 

(2) 

(3) 

(4) (B) and (C) both

37) Which of the followig carbocations do not undergo rearrangement ?

(1) 

(2) 

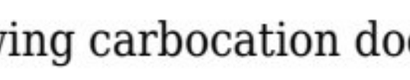
(3)

(4) All of these

38) Arrange the following in correct order of reactivity towards EAR :-



(I)



(II)



(III)

(IV)

(1) III > II > I > IV

(2) IV > III > II > I

(3) II > III > IV > I

(4) I > III > II > IV

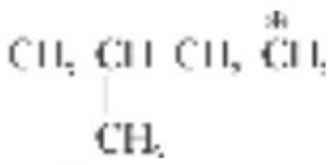
39) Which of the following carbocation does not undergo rearrangement ?

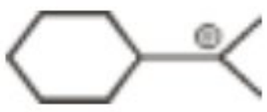
(1)

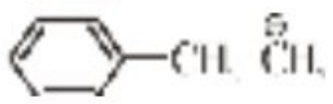
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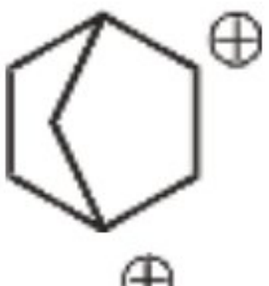
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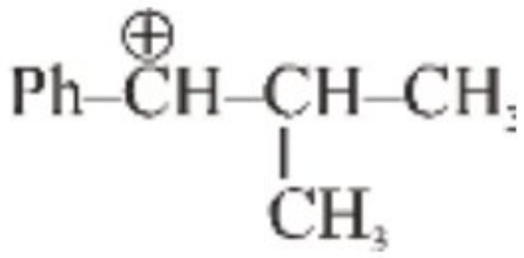
(2) 

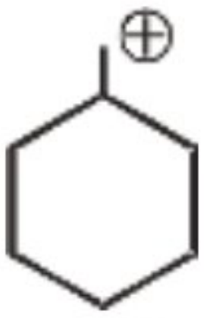
(3) 

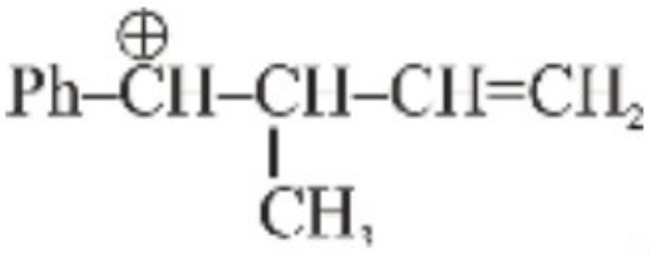
(4) 

40) Which carbocation will rearrange ?

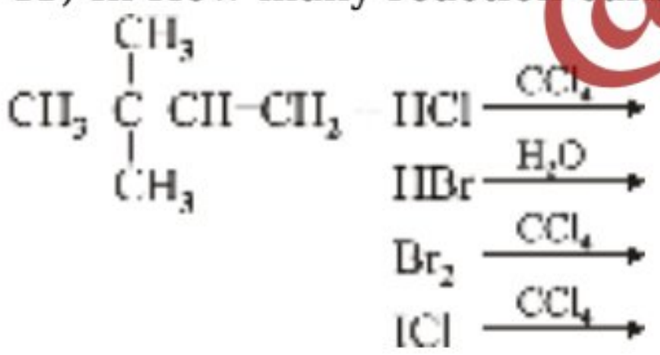
(1) 

(2) 

(3) 

(4) 

41) In How many reaction carbocation rearrangement occur ?



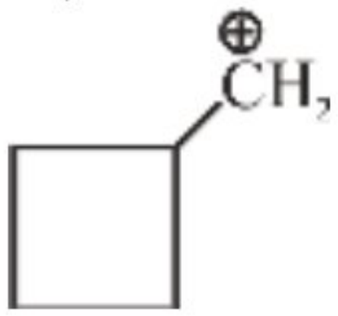
(1) 2

(2) 3

(3) 4

(4) 1

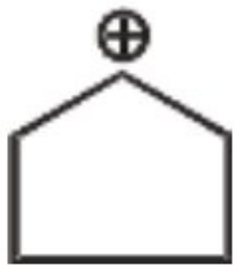
42) Write down final rearranged cation ?

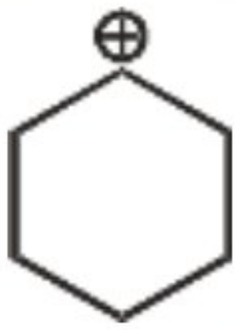


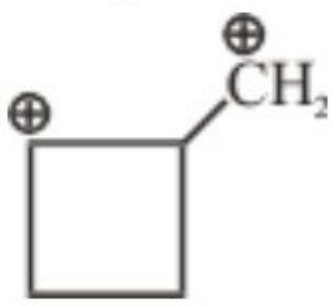
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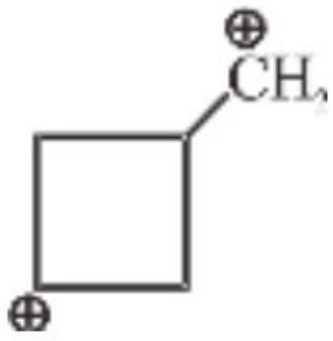
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(1) 

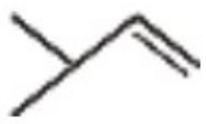
(2) 

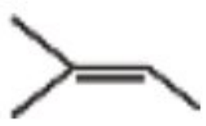
(3) 

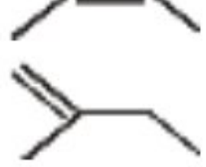
(4) 

43)  ter. pentyl alcohol

A is :-

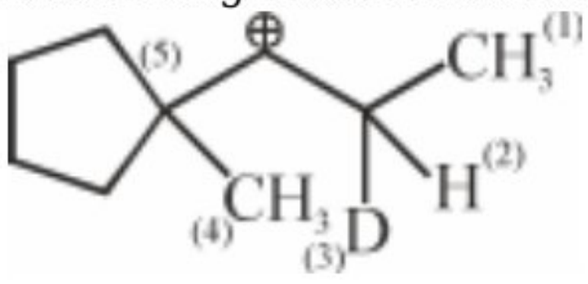
(1) 

(2) 

(3) 

(4) All of these

44) In the following carbocation the most stable rearranged carbocation is formed by migration of group :-



(1) CH_3 (1)

(2) CH_3(4)

(3) C–C bond (5)

(4) D (3)

45) **Assertion (A) :** Alkene is more reactive than alkyne for electrophilic addition reaction.
Reason (R) : Alkyl carbocation is more stable than vinyl carbocation.

(1) Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A).

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(2) Both Assertion (A) and Reason (R) are the true but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) is false and Reason (R) is true.

BIOLOGY

1) Read the following statements.

(A) Traditional hybridisation leads to inclusion and multiplication of undesirable genes along with desired genes.

(B) The first recombinant DNA was constructed in 1972 by linking an antibiotic resistance gene to the native plasmid of E.coli.

(C) More than 900 restriction enzymes have been isolated from over 230 strains of bacteria.

(D) Exonucleases make cuts at specific positions within the DNA.

Choose the statements which are not incorrect.

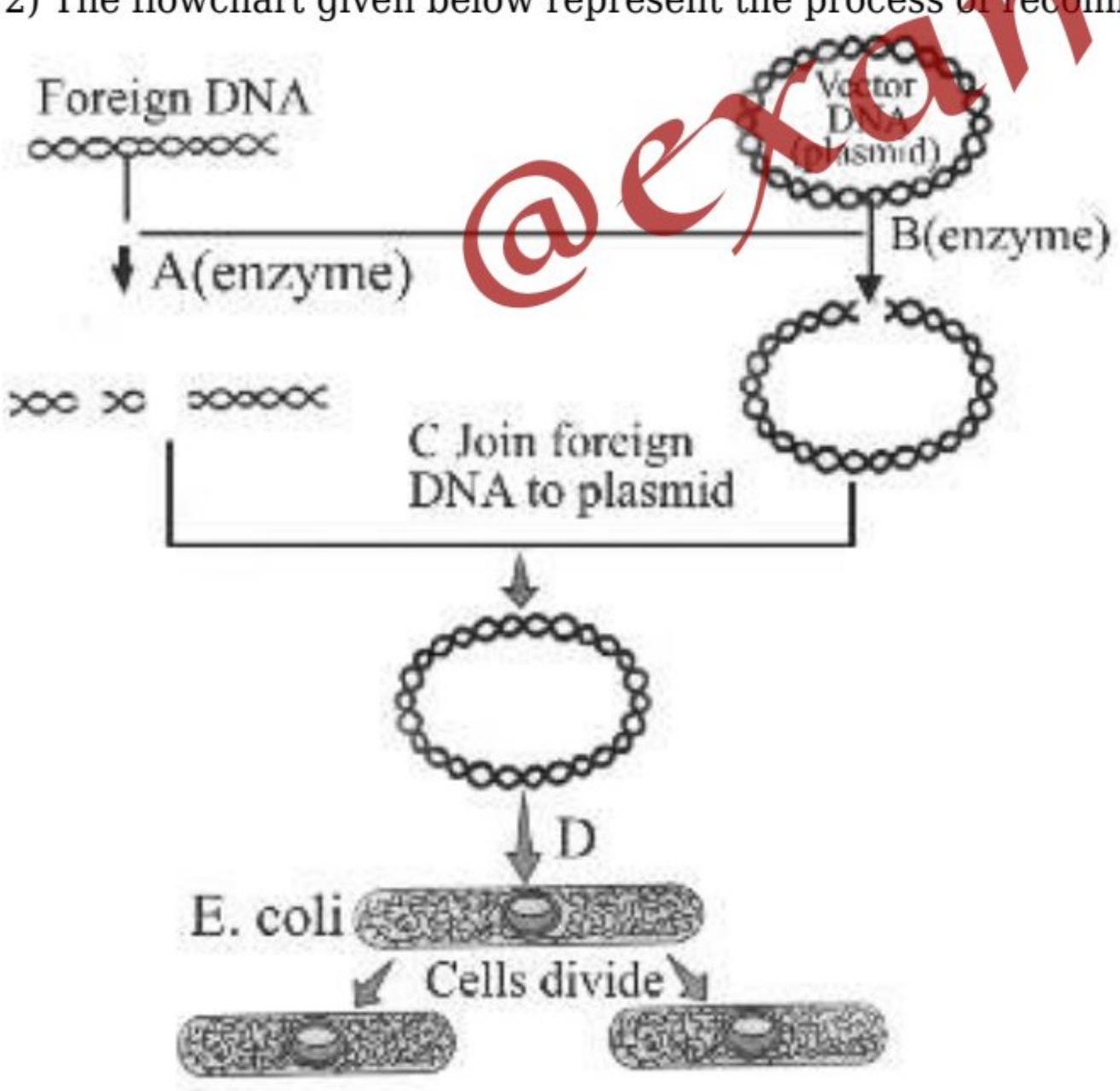
(1) A & B

(2) B & C

(3) B & D

(4) A & C

2) The flowchart given below represent the process of recombinant technology. Identify A to D.



A –Restriction endonuclease,

(1) B–Restriction exonuclease,

C –RNA ligase, D–Transformation

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A –Restritction endonuclease
(2) B–Restriction endonuclease,
C –DNA ligase, D–Transformation

A–Restriction exonuclease
(3) B–Restriction endonuclease,
C –DNA polymerase, D–Transduction

A–Restritction endonuclease,
(4) B–Restriction endonuclease,
C –DNA polymerase, D–Transformation

3) Restriction enzymes belongs to a large class of the enzymes called

(1) Cellulase
(2) Nucleases
(3) Chitinase
(4) Spooling

4) At present about 900 restriction enzymes have been isolated from over 230 strains of bacteria, each of which recognises different recognition sequences. In EcoRI the letter 'R' is derived from the name of

(1) Scientist who discovered the enzyme
(2) Strain of E.Coli
(3) Genus name
(4) Species name

5) **Assertion :** Restriction enzymes recognise palindromic sequences.
Reason : Palindromic sequences read same in both directions of the two strands.

(1) Both assertion and reason are true and reason is the correct explanation.
(2) Both assertion and reason are true but reason is not the correct explanation.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

6) There are two enzymes responsible for restricting the growth of bacteriophage in *E.coli*. Which of the following enzyme prevents the restriction digestion of DNA of *E.coli*?

(1) DNase
(2) Restriction enzymes
(3) Methylase
(4) RNase

7) Which of the following statement is incorrect w.r.t restriction enzymes?

(1) The palindromic nucleotide sequence recognised by *EcoR I* has 6 bp
(2) They cut the strands of DNA at palindromic sites
(3) Sticky ends cannot be joined using DNA ligases

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(4) *EcoR I* produce sticky ends after cutting

8) **Assertion :** Biotechnology work to improve the quality of human life especially in the field of food production and health.
Reason : Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

9)

The first capital letter in the name of restriction enzymes indicates

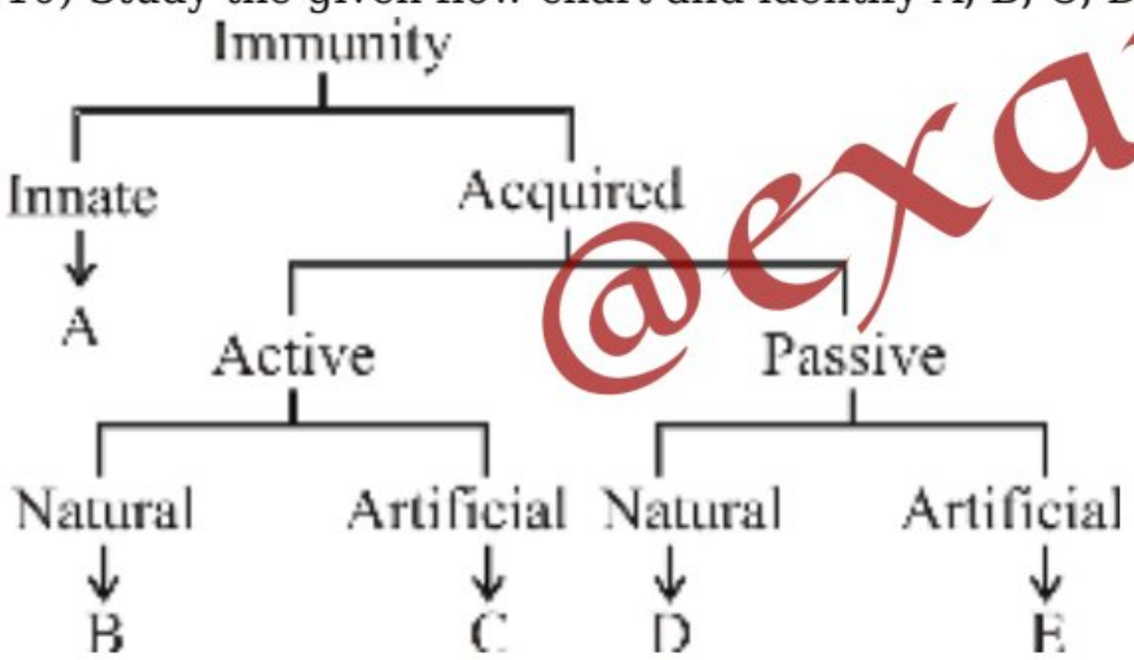
(1) Order in which enzymes were discovered

(2) Order in which enzymes function inside the body

(3) Genus of the prokaryotic cell from which they were isolated

(4) Order in which enzymes were synthesised inside the prokaryotic cell.

10) Study the given flow chart and identify A, B, C, D, E.



	A	B	C	D	E
(A)	Clinical infection	Interferons	Vaccine	Colostrum	Antiserum
(B)	Interferons	Clinical infection	Vaccine	Colostrum	Antiserum
(C)	Interferons	Clinical infection	Colostrum	Vaccine	Antiserum
(D)	Interferons	vaccine	Clinical infection	Sera	Colostrum

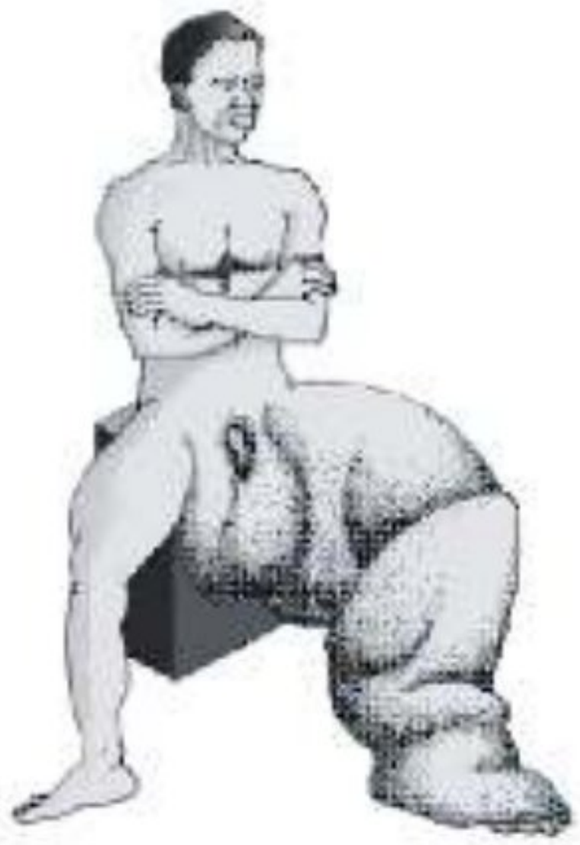
(1) A

(2) B

(3) C

(4) D

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11) Read the following (a-d) -  How many of the following statements are correct about give diagram.
(a) Gross deformities in genital organs.
(b) It is caused by Wuchereria.
(c) It is slowly developing chronic inflammation usually of blood vessels of lower limbs.
(d) Vector is female aedes mosquito.

(1) Four
(2) One
(3) Two
(4) Three

12) Consider the following four statements (a-d) regarding kidney transplant and select the two correct ones out of these.
(a) Even if a kidney transplant is proper the recipient may need to take immuno-suppresants for a longtime.
(b) The cell-mediated immune response is responsible for the graft rejection.
(c) The B-lymphocytes are responsible for rejection of the graft.
(d) The acceptance or rejection of a kidney transplant depends on specific interferons. The two correct statements are :

(1) (a) and (b)
(2) (b) and (c)
(3) (c) and (d)
(4) (a) and (c)

13) Which of the following is correct?

(1) B -cells provide CMI
(2) Interferons prevent the neighbouring cells from further viral infection
(3) As the age increases, thymosin increases
(4) T-cells produces antibodies

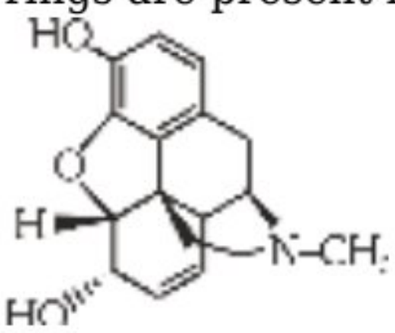
14) (a) Smack (b) Diacetylmorphine
(c) White (d) Odourless
(e) Bitter crystalline compound

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(f) Extracted from latex of poppy plant
Above statements/informations are correct for:-

(1) Morphine
(2) Heroin
(3) Coccain
(4) Barbiturates

15) Diagram is showing chemical structure of Morphine, which is a natural opiate like codeine. Morphine is a strong analgesic, also has sedative & calming effect. It depresses respiratory centre, BP, heart beat. Constipation is a prominant feature of morphine action. How many phenanthrene rings are present in a morphine molecule ?



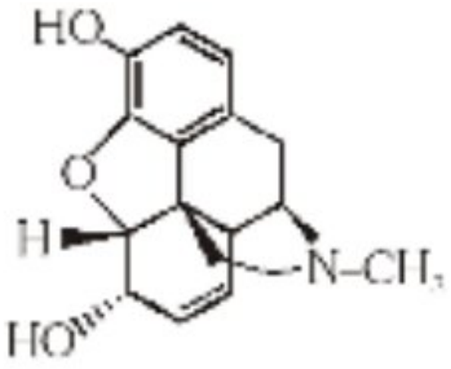
(1) 2
(2) 3
(3) 4
(4) Absent

16) Primary response which is of ____(A)____ intensity. Subsequent encounter with the same pathogen elicit a ___(B)___intensified ___(C)___ response, which is based on ___(D)___.

	(A)	(B)	(C)	(D)
(1)	High	Low	Anamnestic	Specificity
(2)	Low	High	Primary	Diversity
(3)	Low	High	Secondary	Memory
(4)	High	Low	Primary	Memory

(1) 1
(2) 2
(3) 3
(4) 4

17) Diagram is showing chemical structure of Morphine, which is a natural opiate like codeine. Morphine is a strong analgesic, also has sedative & calming effect. It depresses respiratory centre, BP, heart beat. It is obtained from



(1) Cannabis satava
(2) Datura
(3) Erythroxylum coca

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(4) Papever sominiferum

18) Mark the correct statement :-
(a) Yoga has been practised to achieve physical and mental health.
(b) Infectious diseases are very common and everyone of us suffers from these at sometime or other
(c) AIDS is an infectious disease
(d) Cancer is non-infectious disease
(e) Healthy persons bring economic prosperity

(1) a, b and c

(2) b, c, d and e

(3) c and d

(4) a, b, c, d and e

19) Given below are two statements :
Statement-I : When an infected female *Anopheles* mosquito bites, it releases gametocytes of *Plasmodium* into the healthy person.
Statement-II : The female *Anopheles* mosquito takes up sporozoites of *Plasmodium* with blood meal from an infected person, suffering from malaria.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement-I is incorrect but Statement-II is correct

(2) Both Statement-I and Statement-II are correct

(3) Both Statement-I and Statement-II are incorrect

(4) Statement-I is correct but Statement-II is incorrect

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20) Which of the following is not correct about pneumonia ?
(A) It is caused by Streptococcus pneumoniae.
(B) Infects the alveoli of the lungs.
(C) Symptoms of pneumonia include fever, chills, Cough and headache.
(D) In all cases the lips and finger nails turn blue to grayish in colour.

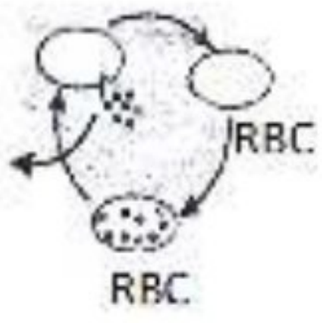
(1) A, B, D

(2) B and D

(3) D only

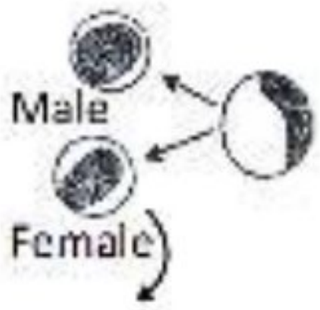
(4) A, B, C, D

21) Make a glance of a part of the life cycle of *Plasmodium* give below as events A, B, C and D along with their description :>

	Events	Description
A		Parasites reproduce asexually in the red blood cells, bursting the red blood cells and causing cycles of fever

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B		Sexual stage (gametocytes) develop in RBCs of mosquito
C		Fertilization and development takes place in GUT of mosquito
D		Mature infective stages (sporozoites) escape from the GUT and migrate to the mosquito salivary glands

How many of the above events are correctly described ?

(1) One

(2) Two

(3) Three

(4) Four

22) Consider the following characteristics and answer the question.

(a) Metastasis

(b) Contact inhibition

(c) Uncontrolled meiosis

(d) Malignancy

(e) Invasion

(f) Oncogenic transformation

The properties shown by malignant tumor are?

(1) a, b, d, e, f

(2) a, d, e, f

(3) a, b, d, e

(4) a, b, c, d, e, f

23) **Assertion:** Crack is a potent stimulant of CNS, producing sense of euphoria and increased energy.

Reason: It interferes with the transport of neurotransmitter dopamine.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

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24) **Assertion:** Monoclonal antibodies are designed to attach specific proteins in cancer cells.
Reason: Immunotherapy helps in early detection of cancer.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

25) Which of the following statement is false ?

(1) The adverse effects are just not restricted to the person who is using drug or alcohol.

(2) At times, a drug/alcohol addict becomes the cause of mental and financial distress to his/her entire family and friends.

(3) Excessive doses of drug may lead to coma and death due to respiratory failure, heart failure or cerebral haemorrhage.

(4) Withdrawal symptoms can never be severe or life threatening.

26) (a) Coke
(b) Diacetylmorphine
(c) Brown in colour
(d) Odourless
(e) Sweet compound
(f) Extracted from latex of Bhang plant
How many statements/informations are correct for heroin :-

(1) Two

(2) Three

(3) Five

(4) Four

27) Which of the following pair is not cannabinoids ?

(1) Charas, Ganja

(2) Datura, LSD

(3) Charas, Hashish

(4) Marijuana, Bhang

28) Mark the incorrect statement :-

(1) The T-cells themselves do not secrete antibodies but help B-cell to produce them

(2) *Wuchereria* is transmitted to a healthy person through the bite of female *Culex* vector

(3) Haemozoin (toxic substance) is associated with Malaria

(4) After getting into the body of the person, HIV releases its ssDNA into macrophages where DNA genome of virus replicates to form viral RNA with the help of enzyme reverse transcriptase

29) Transmission of HIV :- (a) Transfusion of contaminated blood.

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(b) Sexual contact with infected person
(c) From infected mother to her baby through placenta.
(d) Sharing infected needles
(e) Sharing food with infected person
Which of the statements given above is/are not correct ?

(1) a, b, e

(2) e only

(3) a, c, d, e

(4) a, b, c, d

30) Identify correct about allergy
(I) IgE antibodies are produced
(II) Anaphylatic shock is an allergic condition.
(III) Allergy treatment is done by administration of antihistamines.
(IV) AIDS is an allergic disorder.

(1) Only two statements are correct

(2) Only IV statements is correct

(3) Only three statements are correct

(4) All Four statements are correct

31) Match different organisms in column-I with their uses in column-II and select the correct answer from codes given below.

Column-I		**Column-II**	
(A)	*Lactobacilus acidophilus*	(i)	Formation of dough
(B)	*Saccharomyces cerevisiae*	(ii)	Single cell proteins
(C)	*Propionibacterium sharmanii*	(iii)	Conversion of milk into curd
(D)	*Spirulina*	(iv)	Formation of swiss cheese

(1) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)

(2) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)

(3) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

(4) (A)-(i), (B)-(iii), (C)-(ii), (D)-(iv)

32)

Name the blank space A, B, C and D given in the following table :-

Microbe name	**Commerical product**
A	Lactic acid
B	Cyclosporin–A
Monascus purpureus	C
Penicillium notatum	D

(1) A – *Lactobacillus*, B – *Trichoderma*, C – Penicillin, D –Statins

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(2) A – *Trichoderma*, B – *Lactobacillus*, C – Statins, D – Penicillin

(3) A – *Lactobacillus*, B – *Aspergillus*, C – TPA, D – Penicillin

(4) A – *Lactobacillus*, B – *Trichoderma*, C – Statins, D – Penicillin

33)

The fungal symbiont in the mycorrhizal associations -
(a) absorbs phosphorus from soil
(b) provides resistance to stem-born pathogens
(c) provides tolerance to salinity & drought.
(d) causes an overall increase in growth of plant
The correct statements is/are -

(1) (a) & (b)

(2) (a), (c) & (d)

(3) (a), (b) & (d)

(4) (a), (b), (c) & (d)

34)

Match the column-I with column-II.

	Column-I		**Column-II**
(a)	Curd production	(i)	LAB
(b)	Ethanol production	(ii)	Methanogen
(c)	Citric acid	(iii)	Yeast
(d)	Biogas	(iv)	Aspergillus niger

(1) a-i, b-iii, c-iv, d-ii

(2) a-iv, b-iii, c-ii, d-i

(3) a-i, b-iii, c-ii, d-iv

(4) a-iv, b-iii, c-i, d-ii

35) _____ are free-living fungi that are common in the root ecosystem.

(1) *Aspergillus niger*

(2) *Trichoderma*

(3) *Acetobacter aceti*

(4) *Bacillus thuringiensis*

36) Select the correct match -

(1) *Aspergillus niger* - Acetic acid

(2) Ganga action plan - UNEP

(3) Biogas - IARI

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(4) Wine, beer - distillation product

37) Which enzyme produced by microbes is an important ingredient in detergent formulations?

(1) Urease
(2) Lipase
(3) Pectinase
(4) Cleansase

38) BOD can be defined as

(1) Amount of CO_2 consumed if all the inorganic matter in one litre of water is oxidized by bacteria.
(2) Amount of O_2 consumed if all the organic matter in one decilitre of water is reduced by bacteria.
(3) Amount of O_2 consumed if all the organic matter in one litre of water is oxidized by bacteria.
(4) Amount of O_2 consumed if all the inorganic matter in one litre of water is oxidized by bacteria.

39) Methanogens produces which gases amongst the following list of gases?
CH_4, O_2, CO_2, H_2O_2, H_2S, H_2

(1) CH_4, H_2O_2, H_2
(2) H_2S, H_2, CO_2
(3) CH_4, H_2, CO_2
(4) H_2S, H_2O_2, CO_2

40) Which of the following statement is correct :

(1) Microbes are also used in ferment fish soyabean and bamboo shoot.
(2) Roque fort chesse are ripened by algae.
(3) Puffed up appearance of dough is due to O_2
(4) Microbes are not diverse

41) The following diagram represents:



(1) Adenovirus particles which is capable of causing respiratory infections
(2) Proteinaceous infectious agents
(3) A prokaryote which causes respiratory disorder
(4) Fungal colony growing in a petridish

42) **Statement-I :** Methanogens grow aerobically on cellulosic material, produces large amount of methane along with CO_2 and H_2.

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Statement-II : Human beings are not able to digest cellulose.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

43) **Statement-I :** Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria(LAB) grow in milk and convert it to curd.
Statement-II : In our stomach, the LAB play very beneficial role in checking disease causing microbes.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

44) Find the incorrect statement w.r.t Biocontrol agents

(1) The ladybird, and dragon fly are useful to get ride of aphids and mosquitoes respectively.
(2) Bt are available in sachets as dried spores which are mixed with water and sprayed on vulnerable plants such as brassicas and fruit trees.
(3) Trichoderma species are parasitic fungi that are very common in the root ecosystems.
(4) Baculoviruses are excellent candidates for species specific, narrow spectrum insecticidal application.

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45) **Assertion (A):** Statins produced by Monascus purpureus have been commercialized as bloodcholesterol lowering agents
Reason (R): Statins competitively inhibit the enzymes responsible for synthesis of cholesterol

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

46) In a dihybrid test cross involving two linked genes with 18.5% recombination frequency, what percentage of the progeny will exhibit the recombinant phenotype?

(1) 9.25%
(2) 18.5%
(3) 81.5%
(4) 40.75%

47) A normal vision man whose father was colour blind marries a woman whose mother was colour blind and father was normal. What is the probability that their first son will be colour blind?

(1) 0%

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(2) 25%

(3) 50%

(4) 100%

48) In a cross between plants with genotype AaBbCc × AaBbCc (assuming independent assortment for all genes), what proportion of offspring will have the genotype AABbCc?

(1) 1/64

(2) 2/64

(3) 4/64

(4) 8/64

49) Match List-I (Disorders) with List-II (Genetic / Chromosomal Causes) according to NCERT:

	List -I		List-II
(a)	Turner's syndrome	(i)	Autosomal inborn error lacking phenylalanine hydroxylase
(b)	Thalassemia	(ii)	47, XXY karyotype with sterile gynaecomastia
(c)	Phenylketonuria	(iii)	45 with X0 karyotype with sterile rudimentary ovaries
(d)	Klinefelter's syndrome	(iv)	Quantitative autosomal recessive RBC synthesis defect

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

50) In Drosophila melanogaster, T.H. Morgan crossed yellow-bodied (y), white-eyed (w) females with brown-bodied (y+), red-eyed (w+) males. In the F2 generation, the proportion of parental and recombinant phenotypes were respectively:

(1) 62.8% and 37.2%

(2) 98.7% and 1.3%

(3) 50.0% and 50.0%

(4) 81.5% and 18.5%

51) In Antirrhinum majus (Snapdragon), a cross between a pink-flowered plant and a red-flowered plant will produce progeny with which phenotypic ratio?

(1) 1 Red : 2 Pink : 1 White

(2) 1 Red : 1 Pink

(3) 3 Red : 1 Pink

(4) All Pink

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52) Which of the following statements about Sickle-cell anaemia is INCORRECT based on NCERT?

(1) It is caused by the substitution of Glutamic acid by Valine at the 6th position of the β-globin chain.

(2) The single base substitution at codon 6 of mRNA is from GAG to GUG.

(3) Heterozygous individuals (HbA HbS) are clinically diseased under normal oxygen tension.

(4) Under low oxygen tension, the mutant hemoglobin undergoes polymerisation causing shape change.

53) In human ABO blood group system, if both parents have blood group 'AB', how many distinct genotypes and phenotypes are possible among their offspring?

(1) 3 Genotypes, 3 Phenotypes

(2) 4 Genotypes, 3 Phenotypes

(3) 3 Genotypes, 2 Phenotypes

(4) 4 Genotypes, 4 Phenotypes

54) **Assertion (A):** Non-disjunction of the 21st pair of autosomes during oogenesis leads to Down's syndrome in offspring.
Reason (R): Down's syndrome is an aneuploid condition resulting in trisomy of chromosome 21 (47, +21).

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

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55) **Statement I:** In grasshopper, sex determination is XX-XO type where males have only one X-chromosome besides autosomes.
Statement II: In birds, female heterogamety (ZW female and ZZ male) is observed.
Statement III: In honeybees, males develop parthenogenetically from unfertilized eggs and produce sperms by meiosis.

(1) Statements I and II are correct; III is incorrect

(2) Statements I and III are correct; II is incorrect

(3) All Statements I, II, and III are correct

(4) Only Statement II is correct

56) Distance between genes 'A' and 'B' is 12 cM, 'B' and 'C' is 4 cM, and 'A' and 'C' is 8 cM. The correct linear sequence of these genes on the chromosome is:

(1) A – B – C

(2) A – C – B

(3) B – A – C

(4) C – A – B

57) In Pisum sativum, starch synthesis is controlled by gene 'B'. Heterozygotes (Bb) produce

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intermediate-sized starch grains. This phenomenon demonstrates:

(1) Complete dominance for seed shape and incomplete dominance for starch grain size by the same gene.

(2) Codominance for both seed shape and starch grain size.

(3) Multiple allelism with polygenic inheritance.

(4) Epistasis of gene B over seed coat color.

58) Who experimentally verified the 'Chromosomal Theory of Inheritance' by identifying that sexual reproduction produces variations in Drosophila melanogaster?

(1) Walter Sutton and Theodore Boveri

(2) Thomas Hunt Morgan and colleagues

(3) Gregor Johann Mendel

(4) Alfred Sturtevant and Colin Munro MacLeod

59) How many types of genetically distinct gametes can be produced by an individual with the genotype Aa BB Cc Dd ee FF (assuming independent assortment)?

(1) 4

(2) 8

(3) 16

(4) 32

60) Regarding α-Thalassemia and β-Thalassemia, which of the following is correct?

(1) α-Thalassemia is controlled by two closely linked genes HBA_1 and HBA_2 on chromosome 16.

(2) β-Thalassemia is controlled by a single gene HBB located on chromosome 16.

(3) Thalassemia is a qualitative disorder whereas sickle-cell anaemia is a quantitative disorder.

(4) Deletion of a single allele of HBA_1 leads to fatal hydrops fetalis in all cases.

61) How many meiotic and mitotic divisions are respectively required to produce 400 viable seeds in a typical diploid angiospermic plant?

(1) 500 meiotic and 1200 mitotic

(2) 400 meiotic and 1200 mitotic

(3) 500 meiotic and 1600 mitotic

(4) 400 meiotic and 800 mitotic

62) In a mature 7-celled, 8-nucleate female gametophyte of an angiosperm, which of the following cellular components undergoes degenerate mitotic activity to form a single bi-nucleate/fused diploid structure?

(1) Antipodal cells

(2) Central cell

(3) Synergids

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(4) Egg cell

63) Identify the correct sequence of layers of the anther wall from outside to inside, and select the correct functional attribute associated with them:

(1) Epidermis → Endothecium → Middle layers → Tapetum; Tapetum cells possess dense cytoplasm and generally more than one nucleus.

(2) Epidermis → Middle layers → Endothecium → Tapetum; Endothecium performs dehiscence due to fibrous cellulosic bands.

(3) Tapetum → Middle layers → Endothecium → Epidermis; Tapetum undergoes polyploidy by normal cytokinesis

(4) Epidermis → Endothecium → Middle layers → Tapetum; Middle layers are persistent and nourish developing pollen grains.

64) Match List-I with List-II regarding the structural features of angiosperm reproductive organs:

	List- I		List II
(a)	Filiform apparatus	(i)	Undifferentiated sheath enclosing radicle
(b)	Perisperm	(ii)	Single large shield-shaped cotyledon
(c)	Coleorhiza	(iii)	Persistent nucellus in seed
(d)	Scutellum	(iv)	Cellular thickenings at the micropylar tip of synergids

(1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(2) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(3) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(4) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

65) In water hyacinth (Eichhornia) and water lily (Nymphaea), pollination takes place by:

(1) Water currents only (Epihydrophily)

(2) Insects or wind, above water level

(3) Water currents below surface (Hypohydrophily)

(4) Snails and small aquatic animals

66) If a tetraploid female plant (4n) is crossed with a diploid male plant (2n), what will be the respective ploidy levels of the endosperm and zygote?

(1) 5n and 3n

(2) 4n and 3n

(3) 6n and 3n

(4) 3n and 2n

67) Which of the following outbreeding devices prevents BOTH autogamy and geitonogamy?

(1) Self-incompatibility in hermaphrodite flowers

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(2) Monoecious condition (e.g., Castor, Maize)
(3) Dioecious condition (e.g., Papaya)
(4) Dichogamy (Protandry / Protogyny)

68) In Citrus and Mango varieties, polyembryony occurs due to:

(1) Fertilization of multiple synergids by multiple pollen tubes
(2) Proliferation of diploid nucellar cells protruding into embryo sac
(3) Cleavage of the primary zygote into several embryonic masses
(4) Direct development of antipodal cells without fertilization

69) **Assertion (A):** Cleistogamous flowers produce an assured seed-set even in the complete absence of pollinators.
Reason (R): Cleistogamous flowers never open and are invariably autogamous.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

70) **Statement I:** The exine is made up of sporopollenin, which is one of the most resistant organic materials known.
Statement II: The generative cell is small, has dense cytoplasm and a nucleus, and floats in the cytoplasm of the vegetative cell.
Statement III: In over 60% of angiosperms, pollen grains are shed at the 3-celled stage.

(1) Statements I and II are correct; III is incorrect
(2) Statements I and III are correct; II is incorrect
(3) All Statements I, II, and III are correct
(4) Only Statement I is correct

71) Which one of the following statements is INCORRECT regarding seed dormancy and viability according to NCERT?

(1) Lupinus arcticus excavated from Arctic Tundra germinated after an estimated 10,000 years of dormancy
(2) Phoenix dactylifera (Date Palm) seed discovered at King Herod's palace remained viable for 2,000 years.
(3) As seed matures, water content is reduced to 10–15% moisture by mass and metabolic activity slows down.
(4) All seeds lose viability within a few weeks of shedding under natural conditions.

72) In sea-grasses such as Zostera, which adaptation is observed for pollination?

(1) Pollen grains are light, non-sticky, and winged
(2) Pollen grains are long, ribbon-like, and lack exine/have mucilaginous sheath
(3) Flowers produce bright colors and copious nectar

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(4) Female flowers reach the surface of water by long stalks

73) The primary function of the filiform apparatus located in the synergid is:

(1) Absorbing nutrients from the nucellus directly
(2) Guiding the entry of the pollen tube into the synergid
(3) Preventing the entry of more than one pollen tube
(4) Initiating the first division of the zygote

74) In an angiospermic embryo, what is the remnant of the second cotyledon present in the grass family known as?

(1) Epiblast
(2) Scutellum
(3) Coleoptile
(4) Mesocoty

75) In apomictic species of Asteraceae and grasses, seeds are produced without fertilization. What is the genetic consequence of using apomictic hybrid seeds in agriculture?

(1) Segregation of hybrid traits occurs in progeny, reducing yield over generations.
(2) Hybrid characters are preserved without segregation, eliminating the need to buy hybrid seeds every year.
(3) Apomictic seeds show high genetic variability due to crossing over during endomitosis.
(4) Resulting crops are completely male-sterile and require hand-pollination.

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76) Read following four statements and select correct statement

(1) Two nucleotides are linked through 5'-3' phosphodiester linkage to form a dinucleotide
(2) A nitrogenous base is linked to the pentose sugar through a ester linkage to form a nucleoside
(3) The plane of one base pair stacks over the other in double helix. This in addition to H-bonds confers stability of the helical structure
(4) The nitrogenous bases linked to phosphate moiety project from the backbone

77)

Choose the correct symbols :

$\text{RNA} \xrightarrow{a} \text{DNA} \xrightarrow{b} \text{DNA} \xrightarrow{c} \text{mRNA} \xrightarrow{d} \text{Polypeptide}$

(1) a = Replication, b = Transcription, c = Translation, d = Transduction
(2) a = Reverse transcription, b = Translation, c = Transcription, d = Replication
(3) a = Replication, b = transformation, c = Transcription, d = Translation
(4) a = Reverse transcription, b = Replication, c = Transcription, d = Translation

78) DNA being.....is preferred for storage of genetic information but for the transmission of genetic information.....is better :–

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(1) Less stable, DNA
(2) More stable, RNA
(3) Less stable, RNA
(4) More stable, DNA

79) In process of replication deoxyribonucleoside triphosphate

(1) act as substrate
(2) providing energy for polymerisation reaction
(3) acting as an enzyme
(4) both (1) & (2)

80) The pitch of the helix of DNA is :-

(1) 3.4 nm
(2) 0.34 nm
(3) 10^{-9}m
(4) 3.4 A°

81) **Statement-I :** Diploid content of human DNA is 3.3×10^9 bp.
Statement-II : Cytosine is common for both DNA and RNA and thymine is present in DNA.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

82) The semiconservative DNA replication scheme was proposed by:

(1) Watson and crick
(2) Meselson and Stahl
(3) Temin and Baltimore
(4) Jacob and monad

83) If a hybrid DNA molecule labelled with ^{15}N is allowed to replicate twice in normal culture medium, the percentage of hybrid DNA after second replication will be

(1) 50%
(2) 12.5%
(3) 25%
(4) 75%

84) In a given DNA segment, the number of nucleotides of guanine is 75 and those of thymine is 75. The total number of nucleotides in the segment will be

(1) 75

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(2) 750

(3) 225

(4) 300

85) A molecular biologist analysed a nucleic acid sample and found that it contain Nitrogen bases in following proportion A = 20%, T = 15%, C = 35%, G = 30%. This data indicates that —

(1) Nucleic acid in the sample is single stranded DNA

(2) Nucleic acid in the sample is double stranded DNA

(3) Nucleic acid in the sample is double stranded RNA

(4) Nucleic acid in the sample is single stranded RNA

86) Choose the correct statement:-

(1) Haploid content of human DNA is 4.6×10^{-6} bp.

(2) A nitrogenous base is linked to pentose sugar through phosphodiester linkage.

(3) X-ray diffraction data of Maurice of Wilkins and Rosalind Franklin was the basis of Watson and Cricks DNA model.

(4) DNA as an acidic substance was first identified by Watson and Crick.

87) **Assertion (A)** :- DNA is considered to be better genetic material than RNA from most organism.
Reason (R) :- 2' OH group present in DNA makes it labile.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is not correct but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

88) DNA packaging in eukaryotes is carried out with the help of histone. The unit of compaction is nucleosome. Which option is correct about nucleosome ?

(1) Octamer of nucleosome consists of 2 copies of each H_2A, H_2B, H_3 and H_1

(2) In a typical nucleosome 200 bp are found

(3) Linker DNA is attached to H4 histone protein

(4) The nucleosome in chromatin are seen as beads on string structure when viewed under light microscope

89) In a nucleotide H_3PO_4 binds to which carbon atom of pentose sugar :-

(1) Only I^{st} carbon

(2) Only 3rd carbon

(3) Only 5th carbon

(4) Both 3rd and 5th carbon

90) Crick, one of the discoverer of DNA double helical structure, was the man of :-

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(1) Physics
(2) Chemistry
(3) Zoology
(4) Botany

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NOVA

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	3	1	2	2	1	2	1	2	1	1	3	3	2	4	1	1	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	2	1	3	2	3	3	1	1	2	2	3	3	2	3	1	1	4	3	2
Q.	41	42	43	44	45															
A.	1	3	2	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	4	4	1	3	4	2	2	1	1	1	4	3	3	1	3	2	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	2	4	1	2	4	4	3	4	2	2	4	3	3	4	4	1	3	3
Q.	86	87	88	89	90															
A.	1	1	4	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	2	2	2	2	3	3	1	3	B	3	1	2	2	3	3	4	4	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	1	2	4	1	2	4	2	3	2	4	2	1	2	3	2	3	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	1	3	1	2	3	3	1	2	2	3	1	1	1	2	1	2	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	1	1	2	1	3	2	1	1	4	2	2	1	2	3	4	2	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	3	4	1	3	2	2	3	1										

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SOLUTIONS

PHYSICS

1) $\varepsilon = -\frac{d\phi}{dt}$

2) Lenz's Law

3) E = B4rv + B2r2v

4)

Question Explanation:
The question asks to calculate the magnetic flux, using B_0 as the magnetic field.

Concept:
Magnetic flux is given by $\Phi = B_0 A \cos\theta$

Solution:
Angle between the magnetic field and the normal to the surface: $\theta = 30°$
Area of circular surface:
$A = \pi r^2$
$\Phi = B_0 \pi r^2 \cos 30°$
Since
$\cos 30° = \frac{\sqrt{3}}{2}$

Final Answer: $\Phi = \frac{\sqrt{3}}{2} B_0 \pi r^2$

$$J = E/\rho$$

$$\frac{E + E^*}{\eta \rho} = \frac{E - E^*}{\rho}$$

$$E^* = \frac{a}{2} b$$

$$E = \frac{\eta - 1}{\eta + 1} \cdot \frac{ab}{2}$$

5)

6) As shown in figure a strip of length dx is in the loop at a distance x from wire.

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Now flux passing through dx thickness is

$$\phi = \int Bds$$

$$\int_0^{\phi} d\phi = \int_d^{(d+a)} \frac{\mu_0 Ib}{2\pi x} dx$$

Now ϕ is the flux passing through the loop.

$$\phi = \frac{\mu_0 Ib}{2\pi}(\ell n x)^{d+a}$$

$$\phi = \frac{\mu_0 bI}{2\pi} \ell n \frac{(d+a)}{(d)}$$

$$\phi = \frac{\mu_0 b}{2\pi} \ell n \frac{(d+a)}{d} \cdot I_0 e^{\frac{-t}{\tau}}$$

Putting value of I

Induced emf $e = \frac{-d\phi}{dt}$.

$$e = -\frac{\mu_0 b}{2\pi} \ell n \left(\frac{d+a}{d}\right) I_0 e^{-t/\tau} \times \left(\frac{-1}{\tau}\right)$$

$$e = \frac{\mu_0 bI}{2\pi\tau} \ell n \frac{(d+a)}{d}$$

7) Flux: $\phi = B.\pi r^2$

Induced emf : $\varepsilon = \left|\frac{d\phi}{dt}\right| = 2\pi Br\frac{dr}{dt} = 2\pi Brv.$

Induced current: $I = \frac{\varepsilon}{R} = \frac{2\pi Brv}{R}$

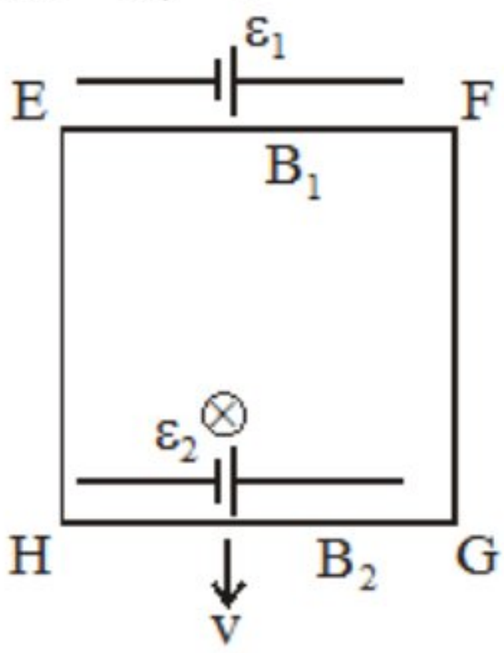
Rate of work done = Rate of heat dissipated in Joule heating = I^2R

$$= \frac{4\pi^2 B^2 r^2 v^2}{R}$$

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8)

$\varepsilon_{EH} = \varepsilon_{FG} = 0$



$\varepsilon = \varepsilon_2 - \varepsilon_1$

$= vB_2a - vB_1a$

$$= va\left[\frac{B_0(y+a)}{a} - \frac{B_0 y}{a}\right]$$

$= vB_0a$

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9) **Solution:**
$B_z \rightarrow$ always remain parallel to surface
$\varepsilon = \pi r^2 \omega$ By sin ωt
Final Answer:
Option (1)

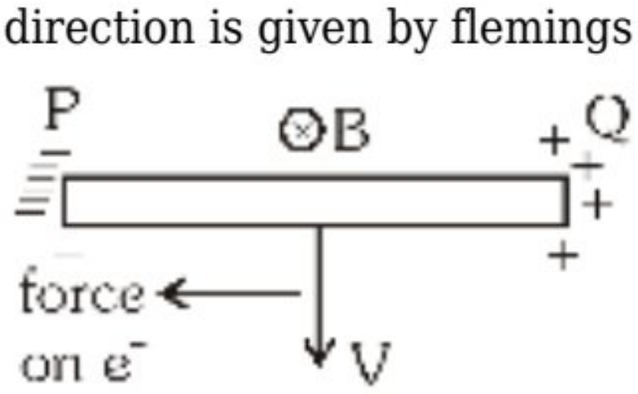
10) **Question Explanation:**
Calculates induced EMF in a rotating rod within a magnetic field.
Concept:
Motional EMF in a rotating conductor.
Solution:
When a conductor rotates in a magnetic field, the induced EMF across a segment of length L from the pivot is given by $V = \frac{1}{2}B\omega L^2$. Using the right-hand rule for the given rotation, the potential at pivot O is higher than at end C.
Calculation:
For segment OC with length
L = 3l: $V_O - V_C = \frac{1}{2}B\omega(3l)^2$
$V_O - V_C = \frac{1}{2}B\omega\left(9l^2\right)$
$V_O - V_C = \frac{9}{2}B\omega l^2$
Final Answer:
Option (2)

11) $\in_{max} = BAN\omega$
$= 0.07 \times 1 \times 10^3 \times 2\pi$
$= 140\pi \approx 440$

12)

induced emf
$= e = B_H lv = 5 \times 10^{-5} \times 3 \times 3$
$= 0.45 \times 10^{-3} = 0.45mV$

13) $e = B \times v \times l_{eff}$
$= B \times v \times 2R$
direction is given by flemings rule



14) Considering a projected length 2R on the ring in vertical plane.
This length will move at a speed v perpendicular to the field. This results in an induced emf:

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e = Bv (2R) in the ring.
e = Bv (2R) in the ring.
In Ring "A" : eA = B (-v) (2R)
In Ring "B" : eB = B (v) (2R)
Therefore, potential difference between A & B = B(v)(2R) - B (iv)(2R)=4BvR.
Note: there will be no p.d. across a diameter due to rotation.
Alternate - Considering rotation of diameter about lowest point:

$e = \frac{B\omega(2r)^2}{2} = 2Bvr$ in A (since pure rotation).

and e = - 2Bvr in B.

15) **Question Explanation:** The question asks to find the total heat energy dissipated in the loop.
Concept: Faraday's Law of Induction and Joule Heating.
Solution:

Magnetic flux linked with the loop:

$$\phi = B \cdot A = \left(B_0 e^{-t/\tau}\right) \cdot (\pi r^2)$$

Induced electromotive force (e):

$$e = -\frac{d\phi}{dt} = \frac{\pi r^2 B_0}{\tau} e^{-t/\tau}$$

Instantaneous power dissipated as heat:

$$P = \frac{e^2}{R} = \frac{\pi^2 r^4 B_0^2}{\tau^2 R} e^{-2t/\tau}$$

Total heat generated from $t = 0$ to $t \to \infty$:

$$H = \int_0^\infty P dt = \frac{\pi^2 r^4 B_0^2}{\tau^2 R} \int_0^\infty e^{-2t/\tau} dt$$

$$H = \frac{\pi^2 r^4 B_0^2}{\tau^2 R} \left[\frac{e^{-2t/\tau}}{-2/\tau}\right]_0^\infty = \frac{\pi^2 r^4 B_0^2}{\tau^2 R} \left(\frac{\tau}{2}\right) = \frac{\pi^2 r^4 B_0^2}{2\tau R}$$

Final Answer: Option (2)

16) **Explanation Question:** We have to calculate the equivalent inductance of the circuit between points A and B
Concept: This question is based on circuit analysis
Solution:
in series
$L_{eq} = L_1 + L_2$ and in parallel

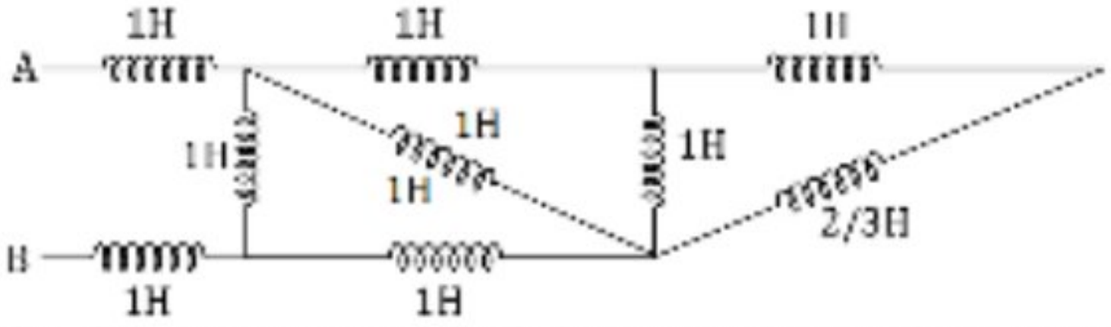
$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}.$$

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Like this way, we can find the equivalent inductance $L_{AB} = 2.618$

Final Answer : The correct option is (4)

17)

Question Explanation:

Two circular coils are arranged in three different relative positions (A), (B), and (C) as shown in the figure. We have to compare the mutual inductance of the two coils in these arrangements and identify the situation in which it is maximum.

Concept:

Mutual inductance dependence on orientation and flux linkage.

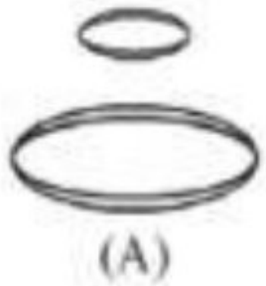
Formula:

$M = \Phi / I$

Solution:

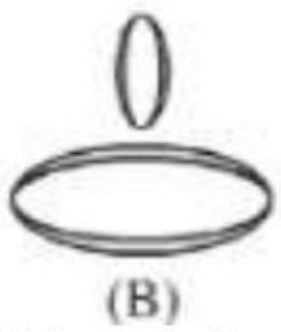
Mutual inductance depends on how much magnetic flux produced by one coil links with the other coil. We have to note that flux linkage is maximum when the area vectors of both coils are parallel and the coils are coaxial.

In situation (A)



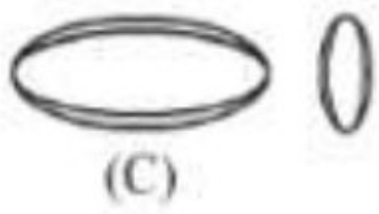
The two coils are coaxial and parallel, so the magnetic field lines produced by one coil pass through the maximum area of the other coil. Hence, flux linkage is maximum.

In situation (B)



Although the centers coincide, the planes of the coils are perpendicular, so the effective flux linkage is reduced.

In situation (C)



The coils are separated sideways, reducing the magnetic flux linking the second coil.

Therefore, mutual inductance is maximum in situation (A).

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Final Answer: Maximum in situation (A)

18) 

$V = 3t^2$

$\frac{Ldi}{dt} = 3t^2$ $\quad u = \frac{1}{2}Li^2$

$\int Ldi = \int 3t^2 dt$ $\quad \frac{du}{dt} = \frac{Li\,di}{dt}$

$Li = \frac{3t^3}{3} + C \Rightarrow \frac{du}{dt} = \left(3t^2\right)\left(\frac{t^3}{L}\right)$

$t=0, i=0 \Rightarrow c=0$

$\Rightarrow i = \frac{t^3}{L}$ $\quad \frac{du}{dt} = \frac{3}{L}t^5$

$t = 1 \quad \frac{du}{dt} = \frac{3}{2} = 1.5\ J/s$

$\Rightarrow x = 1.5$

19) **Question Explanation :** We are given a non-uniform solenoid of length L and a very small radius R(R << L) where the turn density varies linearly with distance from one end as $n(x) = n_0\frac{x}{L}$ for $0 \le x \le L$. The objective is to determine the mathematical expression that the total self-inductance of this setup is proportional to.

Concept: Self-Inductance of a Non-Uniform Solenoid via Integration

Solution: Given the variable turn density $n(x) = n_0\frac{x}{L}$ for a long, thin solenoid (R <<L}, the magnetic field at any position x is:

$B(x) = \mu_0 n(x) I = \mu_0 n_0 \frac{x}{L} I$

The number of turns in an infinitesimal element dx is:

$dN = n(x)dx = n_0\frac{x}{L}dx$

The total flux linkage Λ is found by integrating the flux through dN turns over the entire length L (where cross-sectional area $A = \pi R^2$):

$\Lambda = \int_0^L B(x)AdN = \int_0^L \left(\mu_0 n_0 \frac{x}{L} I\right) A \left(n_0 \frac{x}{L} dx\right)$

$\Lambda = \frac{\mu_0 n_0^2 AI}{L^2}\int_0^L x^2 dx = \frac{\mu_0 n_0^2 AI}{L^2}\left[\frac{L^3}{3}\right] = \frac{1}{3}\mu_0 n_0^2 ALI$

Since self-inductance is defined by $L_{ind} = \frac{\Lambda}{I}$

$L_{ind} = \frac{1}{3}\mu_0 A n_0^2 L \Longrightarrow L_{ind} \propto \frac{1}{3} n_0^2 L$

Final Answer: Option (2)

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20) **Question Explanation :** A thin wire of fixed total length l is initially wound into a long single-layer solenoid of radius R. We need to find the factor by which its self-inductance changes when the same wire is tightly rewound into a new solenoid with a reduced radius of R/2.

Concept: Self-Inductance of a Solenoid (Winding Geometry)

Solution:

Let the original solenoid have N_1 turns, radius $R_1 = R$, length x_1, and cross-sectional area $A_1 = \pi R^2$.

Since the total wire length l is constant, it must satisfy $l = N \cdot (2\pi R)$:

$$N_1(2\pi R) = N_2\left(2\pi\frac{R}{2}\right) \Longrightarrow N_2 = 2N_1$$

Because the wire is tightly packed, the length of the solenoid is determined by the number of turns and the wire diameter \$d\$ (\$x = N \cdot d\$):

$x_2 = 2N_1 \cdot d = 2x_1$

The formula for the self-inductance of a long solenoid is:

$$L = \frac{\mu_0 N^2 A}{x}$$

$$\frac{L_2}{L_1} = \left(\frac{N_2}{N_1}\right)^2 \cdot \left(\frac{A_2}{A_1}\right) \cdot \left(\frac{x_1}{x_2}\right)$$

Substitute the geometric changes $\left(\frac{N_2}{N_1} = 2, \frac{A_2}{A_1} = \frac{1}{4}, \text{and } \frac{x_1}{x_2} = \frac{1}{2}\right)$:

$$\frac{L_2}{L_1} = (2)^2 \cdot \left(\frac{1}{4}\right) \cdot \left(\frac{1}{2}\right) = 4 \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2}$$

Final answer: Option (3)

21) $i = i_0(1 - e^{-Rt/L}) = i_0(1 - e^{-t/T_C})$

$$q = \int_0^{T_C} i\, dt$$

$$= \int_0^{T_C} \frac{\varepsilon}{R}\left(1 - e^{-t/T_C}\right)$$

$$= \frac{\varepsilon}{R}\left(t - \frac{e^{-t/T_C}}{-1/T_C}\right)\Bigg|_0^{T_C}$$

$$= \frac{e}{R}\left(T_C - T_C e^{-1}\right) - \frac{e}{R}(0 + T_C)$$

$$q = \frac{e}{R} \times T_C\ e^{-1}$$

$$= \frac{\varepsilon}{R} \times \frac{L}{R}\ \frac{1}{e}$$

$$= \frac{\varepsilon L}{eR^2}$$

22) $\phi = NBA = LI$

$N\ \mu_0\ nI\pi R^2 = LI$

$N\ \mu_0\ \frac{N}{\ell}\ I\pi R^2 = LI$

N & R constant

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self inductance (L) $\propto \frac{1}{\ell} \propto \frac{1}{\text{length}}$
Option (2)

23) $V_{rms} = \sqrt{\frac{10^2 + 10^2}{2}} = 10V$

24) Due to power losses in transformer,
$P_{output} < P_{input}$ or
$P_{secondary} < P_{primary}$
Hence, power transformation ratio
$\frac{P_{secondary}}{P_{primary}} < 1$

25) **Question Explanation:**
Ideal transformer with given turns and secondary measurements. Find primary voltage and current.

Concept:
Turns ratio for voltages and currents in an ideal transformer.

Solution:
Turns ratio: $\frac{N_p}{N_s} = \frac{500}{2500} = \frac{1}{5}$
For ideal transformer: $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{1}{5}$
$V_p = \frac{1}{5}V_s = \frac{1}{5} \times 200 = 40V$ Also $\frac{I_p}{I_s} = \frac{N_s}{N_p} = 5$
$I_p = 5 \times I_s = 5 \times 8 = 40$ A
Primary: 40 V, 40 A.

Final Answer: Option (2)

26)

Conceptual

27) if R = 0, P = 0

28) Voltage across capacitor will lag current by 90° ;
$i_c = \frac{200\sqrt{2}}{10}\left(\omega t + \frac{\pi}{4} + \frac{\pi}{2}\right)$
$20\sqrt{2}\sin\left(\omega t + \frac{3\pi}{4}\right)$

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29) conceptual

30) Choke limits current without much loss in energy, resistance should be low and inductance should be high.

31)

$i = 5 + 5 \sin(100\ \omega t)$
average value
$< i > = 5 + 5 < \sin(100\ \omega t) >$
In one time period $<\sin(100\omega t) > = 0$
$<i> = 5A$

32)

Explain Question : We need to find rms value representing its effective current.
Concept : This question is based on RMS value for one cycle.

Formula : $I_{rms} = \sqrt{< i^2 >}$
Solution : $i = i_1 \cos\omega t + i_2 \sin\omega t$ square

$$i^2 = i_1^2\cos^2\omega t + i_2^2\sin^2\omega t + 2i_1 i_2 \cos\omega t \sin\omega t$$

Now take average

$$< i^2 > = < \ell_1^2\cos^2\omega t > + < \ell_1^2\sin^2\omega t > + <2i_1i_2\ \sin\omega t\cos\omega t> \ .....(1)$$

$$<\cos^2\omega t> = \frac{1}{2}\ ;\ <\sin^2\omega t> = \frac{1}{2}$$

$<2i_1i_2\ \sin\omega t\cos\omega t> = <i_1i_2\ \sin 2\omega t> = 0$
again from eq. (1)

$$<i^2> = i_1^2\left(\frac{1}{2}\right) + i_2^2\left(\frac{1}{2}\right) + 0$$

$$<i^2> = \frac{i_1^2 + i_2^2}{2}$$

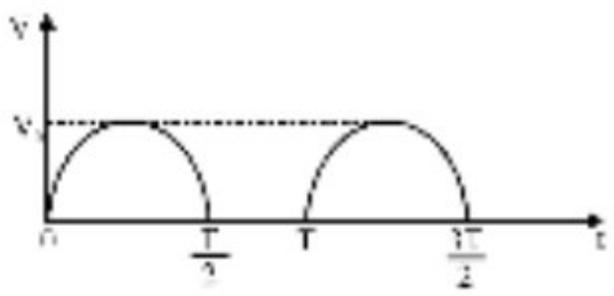
Now take root

$$i_{rms} = \sqrt{< i^2 >} = \sqrt{\frac{i_1^2 + i_2^2}{2}}$$

Final answer : 3

33)

Question Explanation : The waveform of an AC voltage is shown below :



We need to find its root mean square (r.m.s.) value for one full cycle.

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Concept : R.M.S. of an A.C. Voltage

Formula : $V_{rms} = \sqrt{\frac{(V_{rms})_1^2 \times t_1 + (V_{rms})_2^2 \times t_2}{t_1 + t_2}}$

Solution :

$$V_{rms} = \sqrt{\frac{\left(\frac{V_0}{\sqrt{2}}\right)^2 \times \left(\frac{T}{2} - 0\right) + \left(\frac{V_0}{\sqrt{2}}\right)^2 \times \left(T - \frac{T}{2}\right)}{\left(\frac{T}{2} - 0\right) + \left(T - \frac{T}{2}\right)}}$$

$$V_{rms} = \sqrt{\frac{\frac{V_0^2}{2} \times \frac{T}{2} + 0 \times \frac{T}{2}}{\frac{T}{2} + \frac{T}{2}}}$$

$$V_{rms} = \sqrt{\frac{\frac{V_0^2 T}{4} + 0}{T}}$$

$$V_{rms} = \sqrt{\frac{V_0^2}{4}}$$

$$V_{rms} = \frac{V_0}{2}$$

Final Answer : Option (3)

34) **Explain Question:**
Calculate the phase angle by which current leads voltage.

Concept:
Phase angle of series RC circuits.

Solution:
In a series RC circuit, current leads voltage by a phase angle ϕ.
This is determined by the ratio of capacitive reactance X_C to resistance R.

Calculation:

$$* X_C = \frac{1}{2\pi fc} = \frac{1}{2 \times \pi \times 50 \times 100 \times 10^{-6}} = \frac{1}{\pi \times 10^{-2}} = \frac{100}{\pi}\Omega.$$

$$\tan\phi = \frac{X_C}{R} = \frac{100/\pi}{100} = \frac{1}{\pi}$$

$$\phi = \tan^{-1}\left(\frac{1}{\pi}\right).$$

Final answer: Option (2)

35) $\varepsilon_{AB} = 0$

$$\varepsilon_{ABCD} = \frac{d}{dt}(a + bt)\frac{\pi R^2}{2} = \frac{\pi R^2}{2}b$$

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$$\varepsilon_{BC} = \frac{1}{4}\varepsilon_{ABCD} = \frac{\pi R^2 b}{8}$$

36) $B\square v - E - ir = 0$

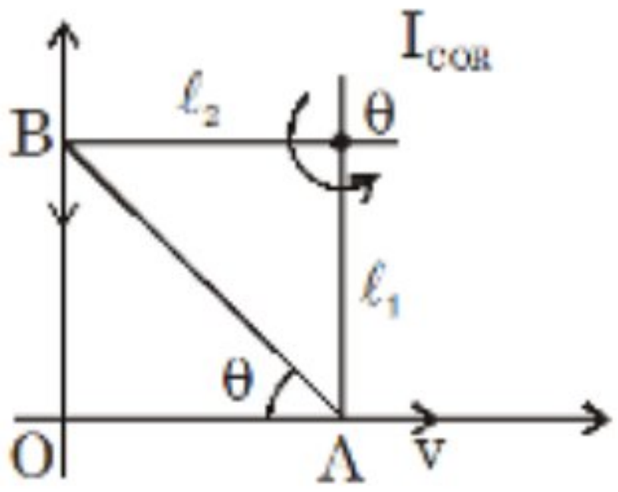
$$F - i\square B = m\frac{dv}{dt} \Rightarrow i = \frac{B\ell v - E}{r}$$

$$P_{avg} = \frac{\int \vec{F}.\vec{v}dt}{\int dt}$$

$$P_{avg} = \frac{\int \left(m\frac{dv}{dt} + i\ell B\right) vdt}{\int dt} = \frac{\int \frac{\ell^2 B^2}{r} v^2 dt}{\int dt}$$

$$= \frac{v^2\ell^2 B^2}{2r} = \frac{4 \times 4 \times 10^{-4}}{2 \times 2 \times 10^{-4}} = 4$$

37)



$$v_B - v_A = \frac{B\omega}{2}\left(\ell_1^2 - \ell_2^2\right)$$

$$0 = \ell_1^2 - \ell_2^2$$

$\square_1 = \square_2$

$\therefore \quad \theta = 45°$

$$\omega_0 = \frac{v}{\ell \sin 45^\circ} = \frac{\sqrt{2}v}{\ell}$$

$$t_0 = \frac{\ell \cos 45^\circ}{v} = \frac{\ell}{\sqrt{2}v}$$

38)
$$I = \frac{\xi}{R_{eq}} = \frac{B\ell v}{\left(\frac{R_{Half}}{2}\right)}$$

39) V developed

$$= \frac{1}{2}B\omega\ell^2 = \frac{1}{2} \times 25 \times 10^{-6} \times \frac{5}{0.25} \times (0.25)^2$$

$= 15.625\ \mu V.$

$$P = \frac{V^2}{R} = \frac{\left(16.625 \times 10^{-6}\right)^2}{0.3 \times 10^{-6}} \approx 0.814mW$$

40) Mutual inductance remains constant.

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41) Flux passing through inner coil at time t .

$$\phi = (\pi a^2)\left[\frac{\mu_0(2t^2)}{2b}\right] = \left(\frac{\pi a^2 \mu_0}{b}\right) t^2$$

$$e = \left|-\frac{d\phi}{dt}\right| = \left(\frac{\pi a^2 \mu_0}{b}\right)(2t)$$

Current flowing through inner coil = $i = \frac{e}{R} = \left(\frac{\pi a^2 \mu_0}{bR}\right)(2t)$

Heat developed from (t) to (t + dt)

$$dH = i^2 R dt = \frac{4\pi^2 a^4 \mu_0^2}{b^2 R} t^2 dt$$

$$\int dH = \frac{4\pi^2 a^4 \mu_0^2}{Rb^2} \int_0^t t^2 dt$$

$$H = \frac{4\mu_0^2 \pi^2 a^4 t^3}{3Rb^2}, k = 3$$

, k = 3

42)

Concept: Average value of AC over a full cycle.

Solution:

The average value of a sinusoidal current over one complete cycle [0,T] is given by $\frac{1}{T}\int_0^T I_m \sin(\omega t)\, dt$.

Since the area under the positive half cycle is equal and opposite to the area under the negative half cycle, the integral over a full period is zero.

Final Answer: Option (3)

43) **Concept:** Heating Effect of AC

Solution:

$$H = \frac{V_{rms}^2}{R} t = \frac{\left(V_m/\sqrt{2}\right)^2}{R} = \frac{V_m^2}{2R} t.$$

If V_m becomes $2V_m$, then $H' \propto (2V_m)^2$

Thus, $H' = 4H$.

Frequency does not affect the heat produced in a pure resistive circuit.

Final Answer: Option (2)

44)

Concept: AC - Current graph

Solution:AC by definition varies periodically, and most common AC is sinusoidal, represented by $i = I_m \sin(\omega t)$. The other shapes do not represent standard alternating current behavior.

Final Answer: Option (3)

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45) **Concept:** Transformer utility and transmission.

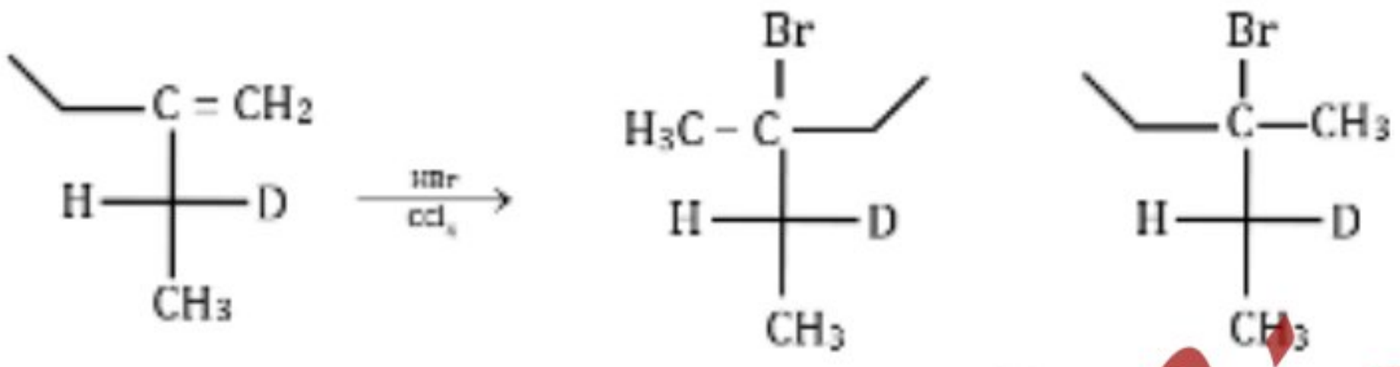
Solution: Though the subtopic is resistive circuits, AC allows high-voltage transmission. High V means low I for the same power, reducing I^2R heat loss in transmission line resistance.

Correct Answer: AC can be stepped up to high voltages to minimize I^2R losses.

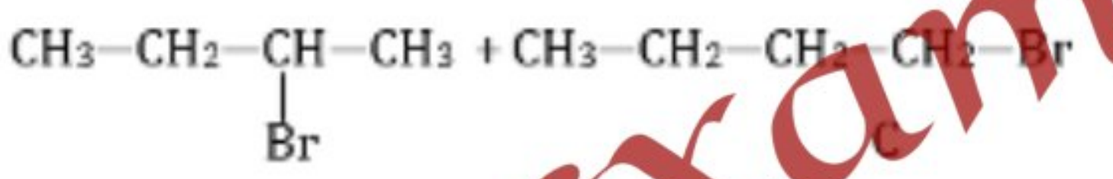
CHEMISTRY

46)

The reaction proceeds via carbocation intermediate which is planar, so $Br^{\ominus}$ has two sides for attack.The products formed are:



Mixture of diastereomers

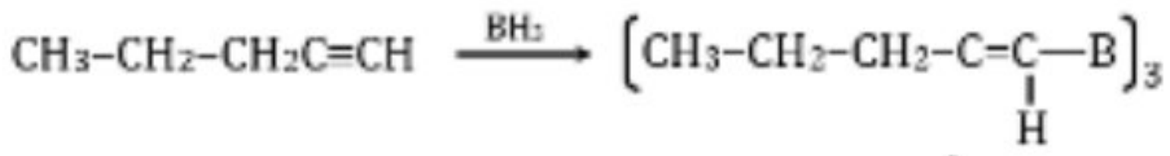


exist in 2 enantiomers (minor product)

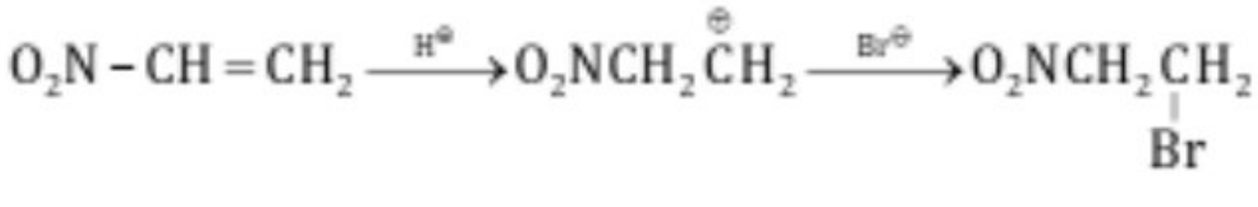
A and B

(Major product)

47)



48)



49)



50)

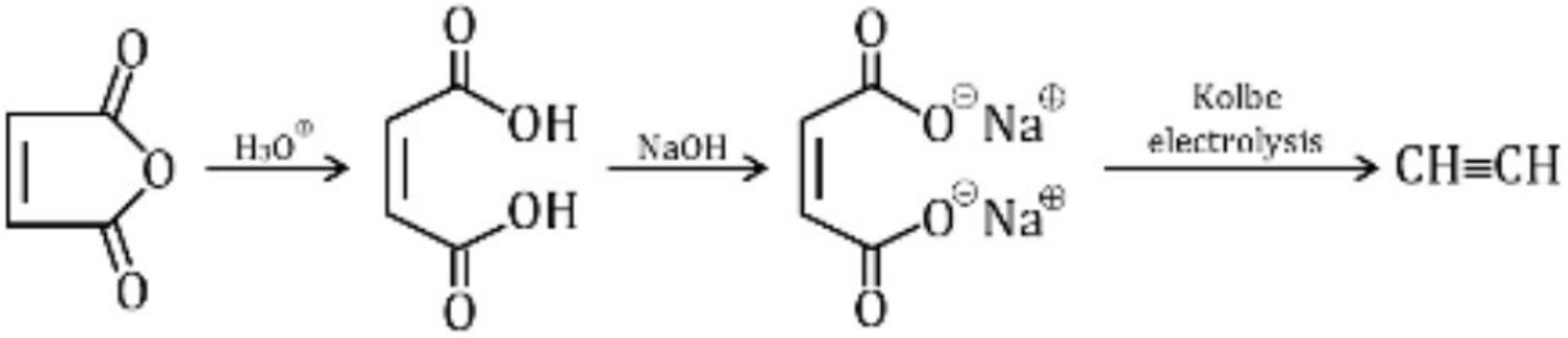
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51) **Question Explanation:**
Determine the final product of a multi-step reaction starting with a cyclic anhydride: hydrolysis to a dicarboxylic acid, salt formation and finally Kolbe electrolysis.

Concept:

This question is based on hydrolysis of anhydrides and Kolbe electrolysis.

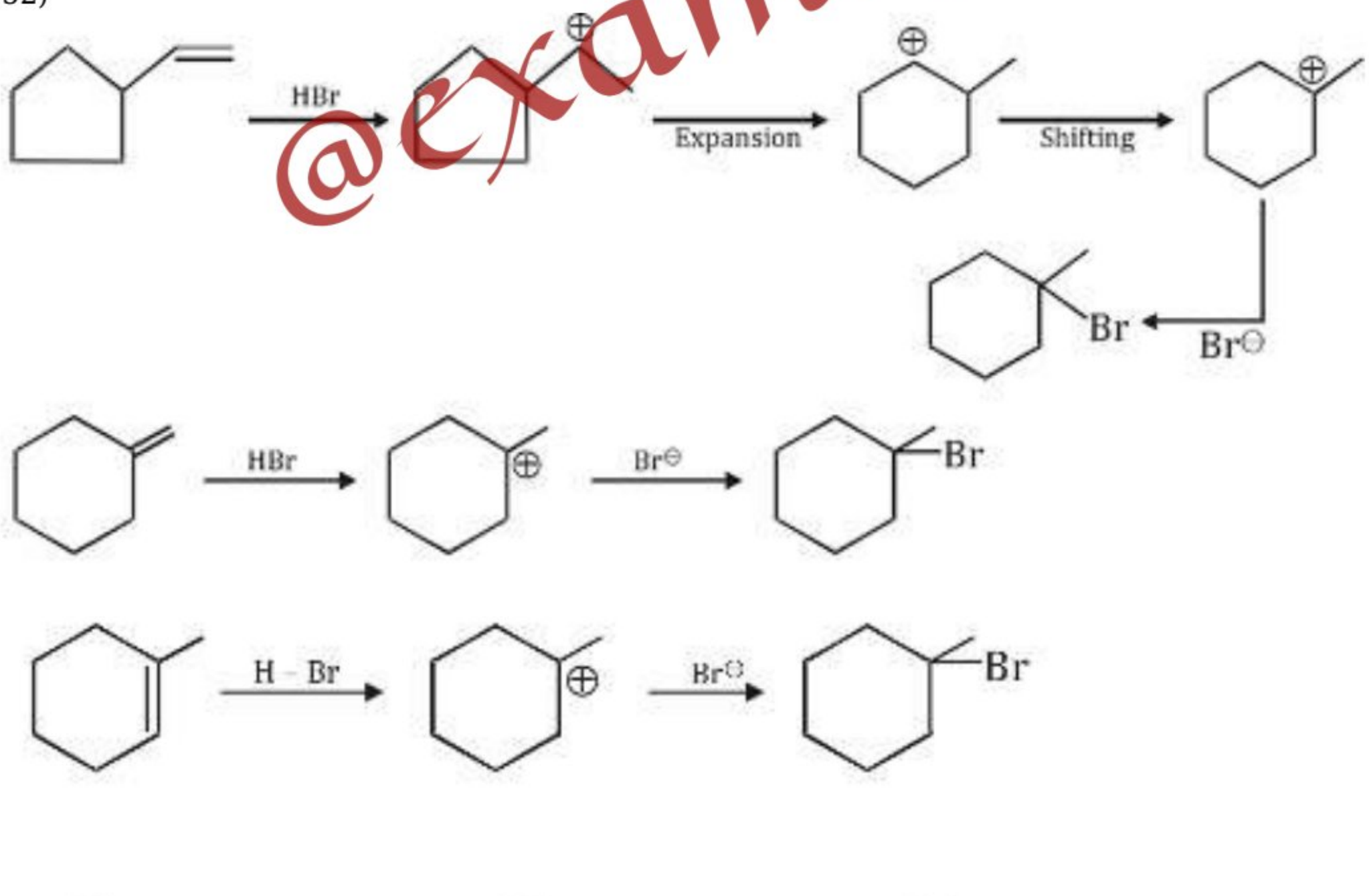
Solution:



Maleic anhydride hydrolyzes with H_3O^+ to form Maleic acid (A). Neutralization with NaOH yields Sodium maleate (B). Kolbe electrolysis of Sodium maleate causes anodic decarboxylation of the dicarboxylate ion, forming a triple bond between the central carbons, resulting in Ethyne.

Final Answer: Option (3)

52)



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53) **Question Explanation:**
Identify the final product formed when propene undergoes hydroboration with B_2D_6 followed by protonolysis (acid cleavage) using CH_3COOH.

Concept:
This question is based on hydroboration-reduction of alkene.

Solution:
Hydroboration (B_2D_6) adds D to the internal carbon and boron to the terminal carbon.
Protonolysis (CH_3COOH) replaces boron on C1 with H, resulting in CH_3-CHD-CH_3.

$$3CH_3-CH=CH_2 \xrightarrow{B_2D_6} (CH_3-\underset{\displaystyle D}{\underset{|}{CH}}-CH_2)_3B \xrightarrow{CH_3COOH} 3CH_3-\underset{\displaystyle D}{\underset{|}{CH}}-CH_3$$

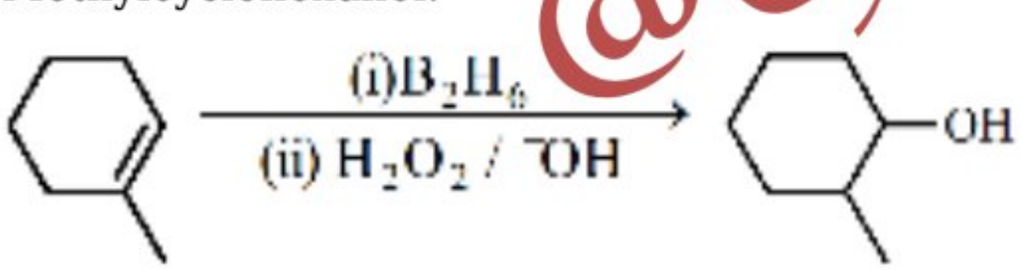
Final Answer: Option (2)

54) **Question Explanation:**
The question asks for the product of the Hydroboration-Oxidation reaction of 1-Methylcyclohexene.

Concept:

This question is based on Hydroboration-Oxidation (Anti-Markovnikov addition).

Solution:
Hydroboration-Oxidation results in the anti-Markovnikov addition of water across the alkene double bond, where the bulky Boron adds to the less hindered carbon.
This is followed by oxidation with H_2O_2/NaOH to yield the final alcohol product, trans-2-Methylcyclohexanol.



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Final Answer: Option (2)

55) **Question Explanation:**
Identify the initial electrophile adding to propene's double bond when reacting with HBr under standard (non-peroxide) conditions.

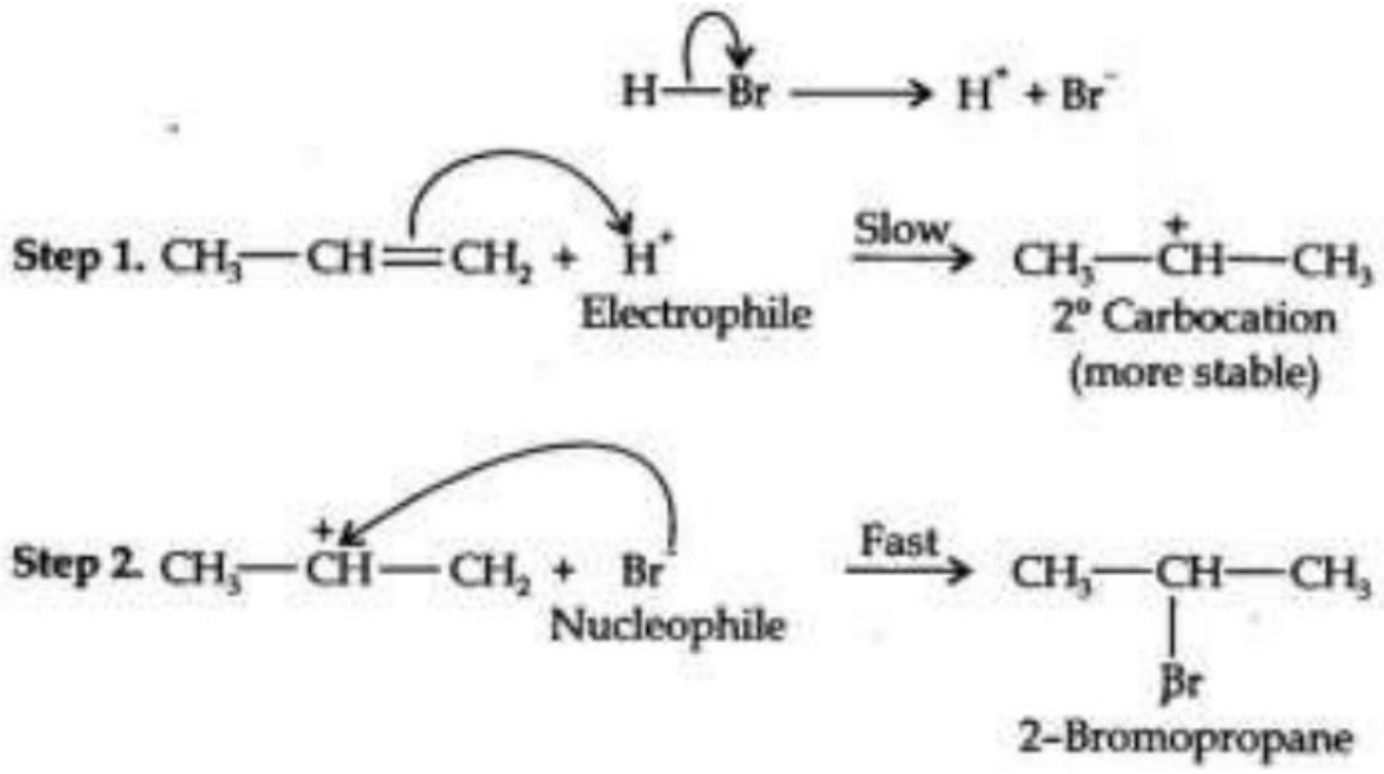
Concept:

This is based on the electrophilic addition mechanism.

Solution:
In the absence of peroxide, the addition of HBr to an alkene proceeds via an ionic electrophilic addition mechanism where the electron-rich double bond attacks the H+ electrophile to form a stable secondary carbocation:

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NOVA



Final Answer: Option (1)

56) **Question Explanation:**
The question asks for the major product of HCl's electrophilic addition to 1-phenylpropene.

Concept:

This question is based on electrophilic addition (Markovnikov's rule).

Solution:
The electrophilic addition mechanism involves the alkene attacking the proton (H^+) to form a stable benzylic carbocation, followed by the chloride ion (Cl^-) attacking the benzylic carbon in a nucleophilic attack.

$$C_6H_5 - CH = CH - CH_3 + H^+ \longrightarrow C_6H_5 - \overset{+}{C}H - CH_2CH_3 \xrightarrow{Cl^-} C_6H_5 - \underset{Cl}{CH} - CH_2CH_3$$

Final Answer: Option (1)

57) **Question Explanation:**
The question requires arranging alkyl free radicals (tertiary, secondary, primary, and methyl) in decreasing order of stability, which dictates their ease of formation.

Concept:

This is based on stability of free radicals, hyperconjugation & inductive effects.

Solution:
The ease of formation of a free radical is directly proportional to its stability. The stability of alkyl free radicals is governed by hyperconjugation (alpha-H) and the +I effect of alkyl groups.

A. Tertiary: Most stable (highest number of alpha-hydrogens and +I groups).
B. Methyl: Least stable (no alpha-hydrogens or +I groups).
Therefore, the order of stability and ease of formation is:

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3° > 2° > 1° > methyl radicals

Final Answer: Option (1)

58)

Question Explanation: Identify the two halides involved in the Wurtz-Fittig reaction.

Concept: Aryl halide + alkyl halide coupling

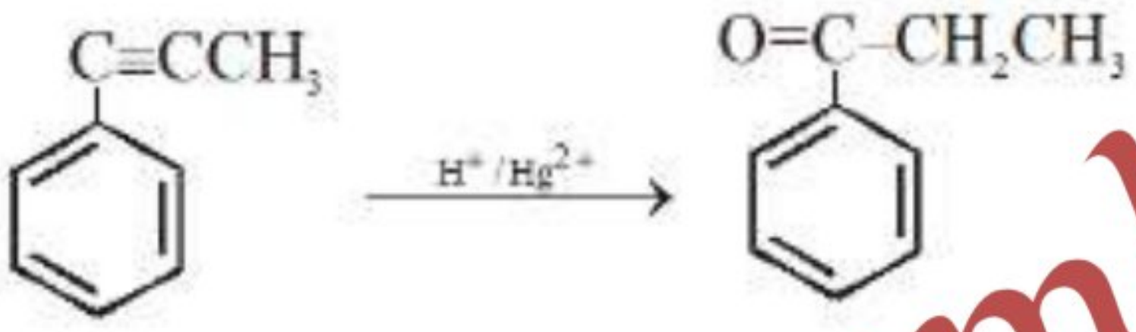
Solution:
Wurtz-Fittig reaction:

$$Ar - X + R - X + 2Na \rightarrow Ar - R + 2NaX$$

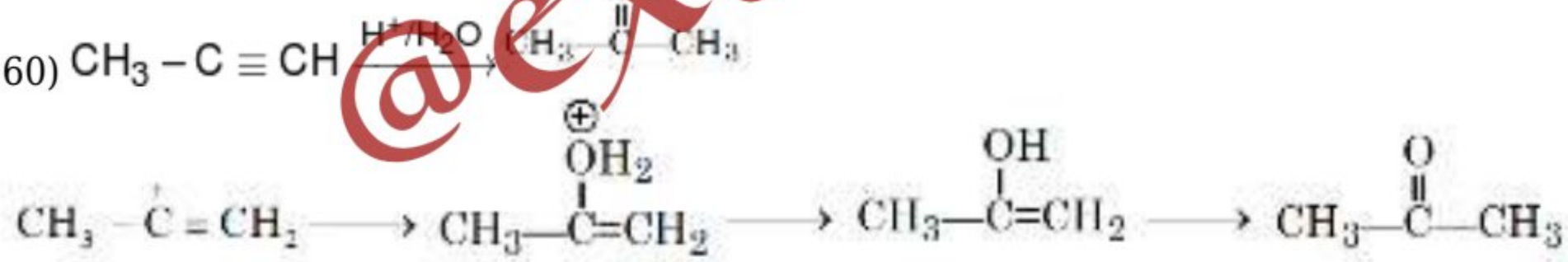
It involves condensation/coupling of one aryl halide and one alkyl halide.

Final Answer: Correct option is (4)

59)

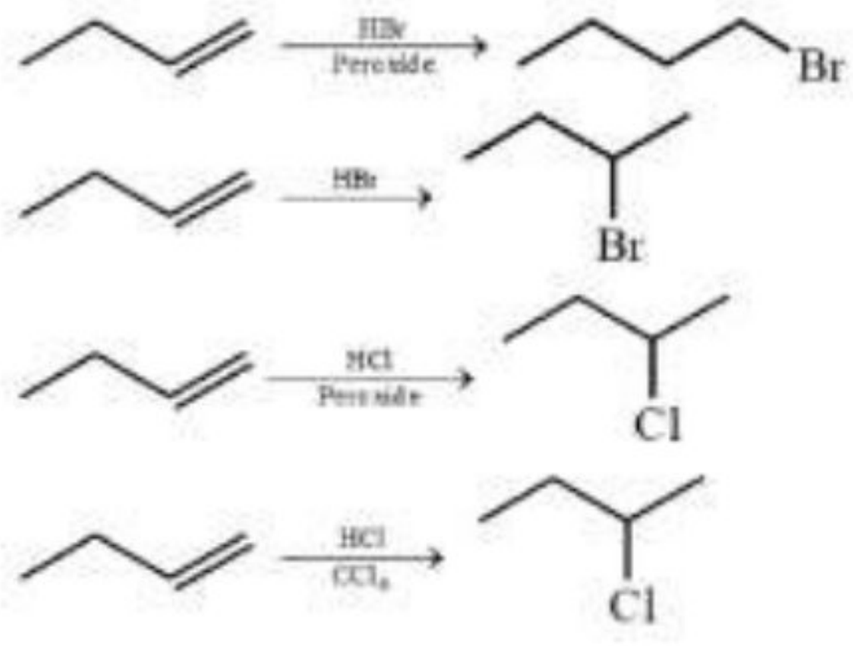


60)



61)

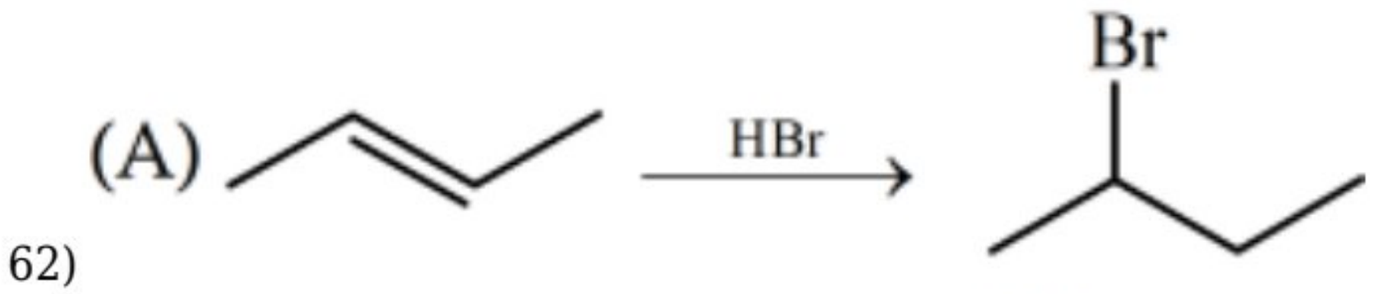
Solution :

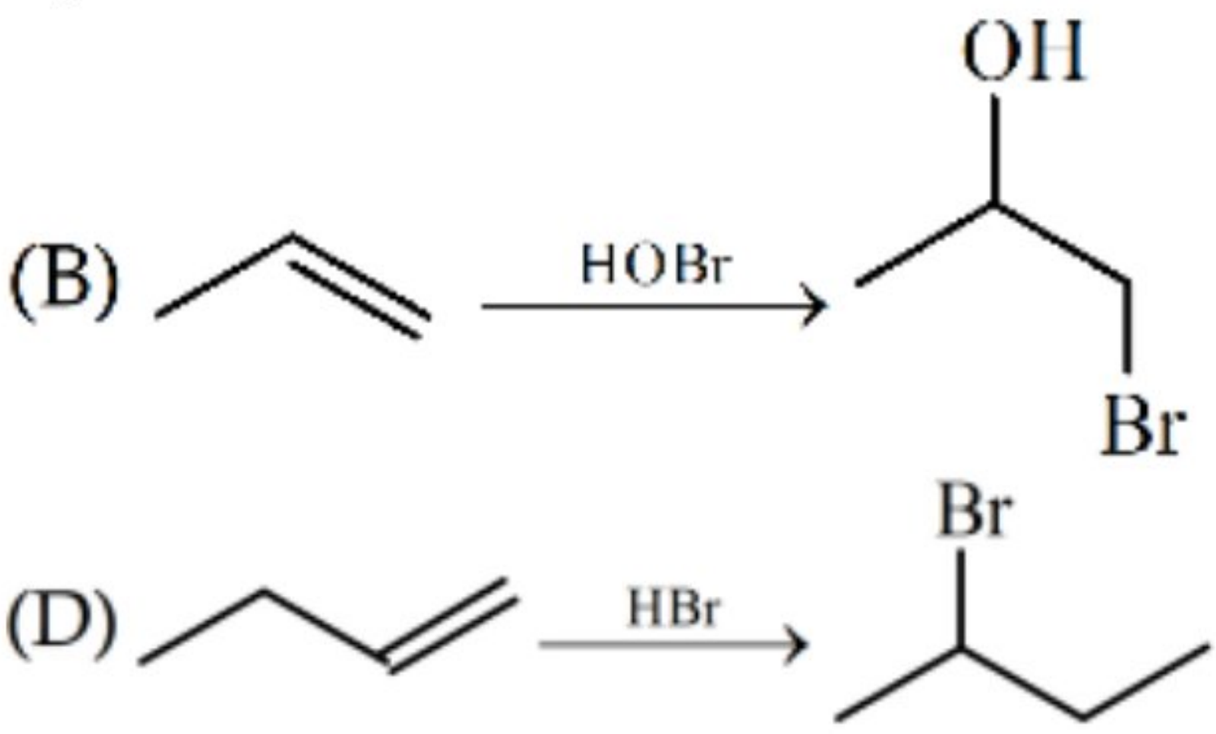


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NOVA

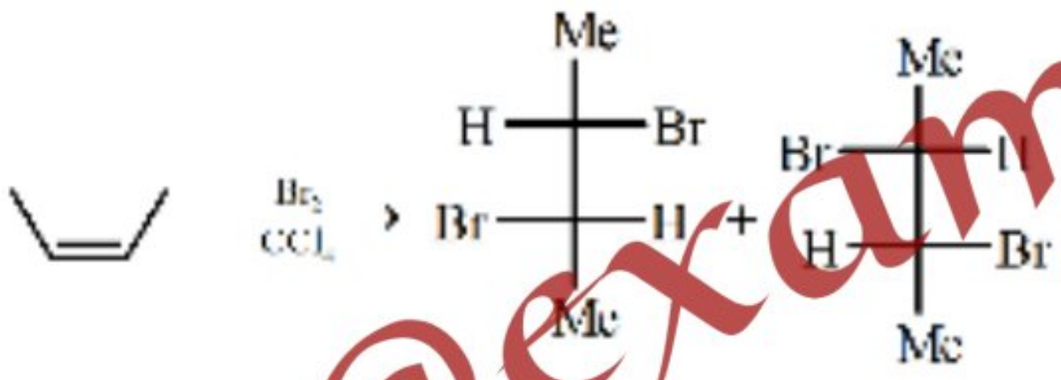
62)





63) Hydrobocation oxidation gives Anti-markovnikov Addition product.

64)



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65)

Hydrobocation oxidation is syn addition which gives anti markownikoff product.

67)

Question Explanation:
The question involves the alkoxymercuration–demercuration reaction of an alkene using methanol.

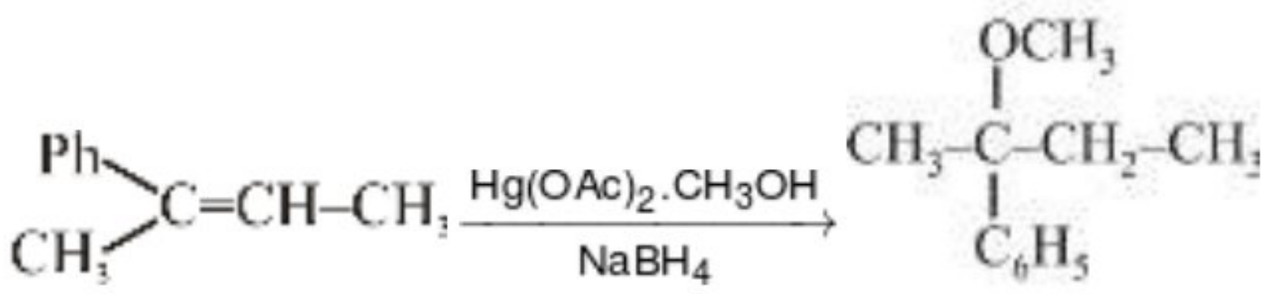
Concept:
This reaction is based on alkoxymercuration–demercuration, a variant of oxymercuration in which an alcohol acts as the solvent and nucleophile.

Solution:
Alkoxymercuration–demercuration proceeds via **Markovnikov addition without carbocation rearrangement**. When **CH_3OH** is used instead of water, the **alkoxy group (–OCH_3)** adds to the **more substituted carbon** of the alkene, while **hydrogen** adds to the **less substituted carbon**.

Based on this regioselectivity, the correct product corresponds to **Option (3)**.

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Final Answer: Option (3)

70)

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71)

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73)

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75)

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76)

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77)

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78)

Conceptual

79) **Question Explanation:** Hydrobromination of 1-ethyl-2-methylcyclohex-1-ene to identify the major product via the most stable carbocation intermediate.

Concept: Carbocation stability, Hyperconjugation, a-hydrogens, Electrophilic addition, Markovnikov's rule.

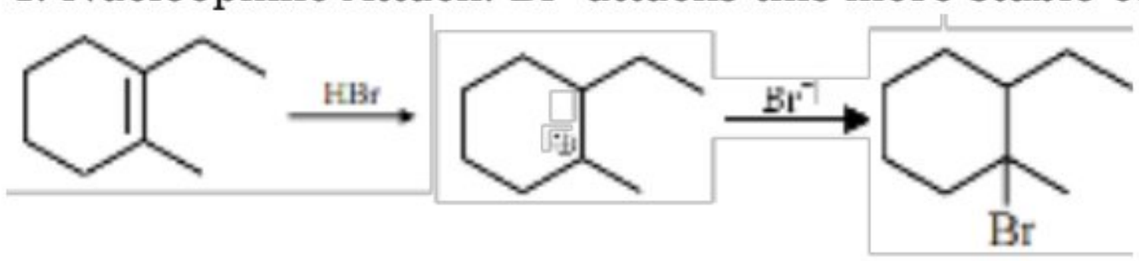
Solution:
1. Protonation: H^+ adds to the ethyl-bearing carbon.
2. Intermediate: This forms a tertiary carbocation on the carbon attached to the methyl group.
3. Stability: This cation has 6 α-H (3 from methyl + 4 from ring CH_2 groups), making it more

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stable than the alternative (5 α-H).
4. Nucleophilic Attack: Br^- attacks this more stable carbocation.



Final Answer: Option (3)

82)

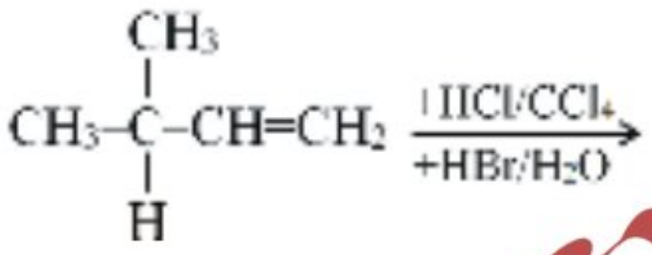
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84)

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85)

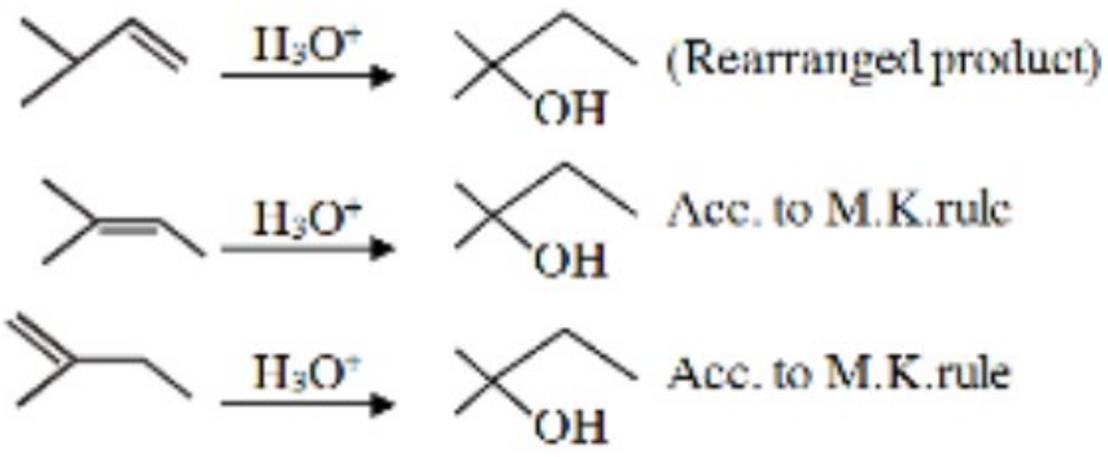
Conceptual

86) 

Carbocation is formed so rearrangement takes place.

87)

Conceptual

88) 

89)

Conceptual

90) **Question Explanation:** The problem asks to verify if alkenes are more reactive than alkynes toward electrophilic addition and if this is due to the relative stability of their carbocation intermediates.

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Concept: Electrophilic addition, Carbocation stability, sp vs sp^2 hybridization, Electronegativity.

Solution:

1. Alkene Reaction: $R - CH = CH_2 + H^+ \rightarrow R - \overset{+}{C}H - CH_3$ (Alkyl carbocation).

2. Alkyne Reaction: $R - C = CH + H^+ \rightarrow R - \overset{+}{C} = CH_2$ (Vinylic carbocation).

3. Stability: Alkyl carbocations are more stable than vinylic carbocations. In vinyl, the positive charge is on a more electronegative sp carbon, which holds electrons tightly and resists positive charge.

4. Reactivity: Higher intermediate stability leads to a lower activation energy, making alkenes more reactive.

Final Answer: Option (1) (Both Assertion and Reason are true, and Reason is the correct explanation).

BIOLOGY

91)

Explain Question: The question is asking to identify which of the given statements are correct (i.e., "not incorrect") and then select the option that contains only those correct statements.

Concept : Traditional hybridisation ,first recombinant DNA and Restriction enzymes

Solution:

Explanation :(A) Traditional hybridisation leads to inclusion and multiplication of undesirable genes along with desired genes: Correct. In traditional plant and animal breeding, entire genomes are crossed. This often results in the inheritance of unwanted traits alongside the specific trait the breeder is trying to select.

(B) The first recombinant DNA was constructed in 1972 by linking an antibiotic resistance gene to the native plasmid of E.coli: Incorrect. While the first recombinant DNA was created in 1972 by Stanley Cohen and Herbert Boyer, they linked the antibiotic resistance gene to the native plasmid of the bacterium Salmonella typhimurium, not E.coli.

(C) More than 900 restriction enzymes have been isolated from over 230 strains of bacteria: Correct. Each of these enzymes recognizes different recognition sequences.

(D) Exonucleases make cuts at specific positions within the DNA: Incorrect. Endonucleases make cuts at specific positions within the DNA. Exonucleases remove nucleotides from the ends of the DNA molecule.

The statements that are not incorrect are A and C.

Final answer: Option 4

96) NCERT page- 170

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97)

Explain Question: The question is asking to identify which statement about restriction enzymes is wrong (incorrect).

Concept: Restriction enzymes

Solution:

Explanation:

The palindromic nucleotide sequence recognised by EcoRI has 6 bp → Correct
(EcoRI recognizes: 5′-GAATTC-3′)

They cut the strands of DNA at palindromic sites → Correct

Sticky ends cannot be joined using DNA ligases → Incorrect
Sticky ends can be easily joined by DNA ligase due to complementary base pairing.

EcoRI produces sticky ends after cutting → Correct

Final answer: Option 3

98)

Question Explanation: We need to analyze the assertion and reason about biotechnology applications.

Concept: Biotechnology Applications

Solution: 1. Assertion (True): Biotechnology works to improve the quality of human life, especially in food production and health. ○ Examples: Genetically modified crops, biofortified foods, vaccines, insulin production.
2. Reason (True): Providing the best catalyst in the form of improved organisms, usually microbes or pure enzymes. ○ Examples: E. coli for insulin production, restriction enzymes in genetic engineering, amylase in food processing.
3. Microbes and enzymes act as key tools to carry out biotechnological processes efficiently.
4. Therefore, the reason directly explains how biotechnology improves food production and healthcare.

Final answer: Both assertion & reason are true & the reason is a correct explanation of the assertion.

99)

The first capital letter in the name of restriction enzyme indicate genus of the prokaryotic cell from which they were isolated.

100)

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101)

Two

104)

NCERT-XII Pg#143

105)

Conceptual

107)

Conceptual

108)

NCERT-XII Pg. # 145-163

109)

Explanation: This question asks which stages of Plasmodium are involved in transmission and whether the given statements are correct.
Concept: Malaria – Life Cycle
Solution: The correct answer is option (3).

A. Statement-I is incorrect: When an infected female Anopheles mosquito bites a human, it releases sporozoites, not gametocytes, into the bloodstream.

B. Statement-II is incorrect: The mosquito does not take up sporozoites; it takes up gametocytes from an infected human.

Final Answer: Both Statement-I and Statement-II are incorrect

110)

Explanation :
(A) It is caused by Streptococcus pneumoniae: This is correct. Pneumonia is commonly caused by Streptococcus pneumoniae along with other bacteria, viruses, or fungi.

(B) Infects the alveoli of the lungs: This is correct. Pneumonia does infect the alveoli, leading to inflammation and fluid buildup.

(C) Symptoms of pneumonia include fever, chills, cough, and headache: This is correct. These are typical symptoms of pneumonia.

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(D) In all cases, the lips and fingernails turn blue to grayish in color: This is incorrect. While cyanosis (blue/gray coloring) may occur in severe cases due to reduced oxygen levels, it is not a universal symptom of pneumonia.
Final Answer : (3)

113) **Explanation:**
The assertion is correct because crack cocaine is a potent CNS stimulant, which leads to euphoria and increased energy.

The reason is also true because crack interferes with dopamine transport, which enhances dopamine levels in the brain, leading to the euphoric effects. This mechanism explains the assertion.

The correct answer is 1. If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

114) **Explanation : Assertion:** Monoclonal antibodies are designed to attach specific proteins in cancer cells. **Reason:** Immunotherapy helps in early detection of cancer.
CORRECT OPTION (2) : If both the assertion and reason are true but the reason: is not a correct explanation of the assertion.

115)

Explaining the Question :- Which statement about drug/alcohol abuse is false ?
Concept:- This question is based on Effects of drug and alcohol abuse

Solution :-
• Statement 4 is false
Withdrawal symptoms can be severe and even life-threatening, especially with substances like alcohol and opioids (e.g., delirium tremens, seizures)
All other statements correctly describe the harmful effects of drug/alcohol abuse.
Final Answer :- 4

116)

Question explanation: This question asks which statements correctly describe heroin (diacetylmorphine).

Concept: Narcotic drugs.

Solution: The correct answer is Option 1 – Two.

A. (a) Coke → (Coke refers to cocaine, not heroin)

B. **(b) Diacetylmorphine** → (This is the correct chemical name of heroin)

C. (c) Brown in colour → (Pure heroin is white)

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D. **(d) Odourless** → (Heroin is odorless)

E. (e) Sweet compound → (Heroin is bitter)

F. (f) Extracted from latex of Bhang plant → (Bhang/cannabis is unrelated; heroin is from opium)

Correct statements: (b) and (d) → 2 statements

Final answer: Option (1)

117)

Question explanation: Identify the pair that **does not belong to cannabinoids**.
Concept: This question is based on Drug abuse
Solution: The correct answer is **option 2**.

A. **Cannabinoids** are compounds derived from **Cannabis plant** → examples: Charas, Ganja, Hashish, Marijuana, Bhang.

B. **Datura** contains **tropane alkaloids**, not cannabinoids.

C. **LSD (Lysergic acid diethylamide)** is a **synthetic hallucinogen**, not a cannabinoid.

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Not cannabinoids → Datura, LSD

Final answer: Option 2

118)

Explain Question : To identify the incorrect statements

Concept : This question is based on Acquired immunity, aids, diseases.

Solution :
After getting into the body of the person, HIV releases its ssDNA into macrophages where DNA genome of virus replicates to form viral RNA with the help of enzyme reverse transcriptase is incorrect
HIV releases ssRNA, not ssDNA. Reverse transcriptase converts ssRNA → dsDNA, which integrates into host genome

Final Answer : option (4)

120)

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Question explanation : Check which statements about allergy are correct.

Concept: Immunology – Allergy

Solution:

A. (I) IgE antibodies are produced → Correct. Allergy involves IgE-mediated hypersensitivity.
B. (II) Anaphylactic shock is an allergic condition → Correct. Severe systemic allergic reaction due to massive histamine release.
C. (III) Allergy treatment is done by administration of antihistamines → Correct. Antihistamines block histamine receptors to reduce symptoms.
D. (IV) AIDS is an allergic disorder → Incorrect. AIDS is caused by HIV infection, not an allergy.

Correct statements = I, II, III.

Final Answer: Option 3

124)

Explain Question: The question asks about matching microbial products with the organisms responsible for their production.

Concept: Intra microbes in household products,industrial product and bio gas

Solution:
(a) Curd production → LAB → (i)
(b) Ethanol production → Yeast → (iii)
(c) Citric acid → Aspergillus niger → (iv)
(d) Biogas → Methanogen → (ii)
Thus, the correct matching is a-i, b-iii, c-iv, d-ii.

Final answer: Option 1

127)

NCERT pg.no.183

130) NCERT - XII, Pg. # 151

132)

Conceptual

133)

Both Statement-I and Statement-II are correct.

135)

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Conceptual

136) Conceptual

137) Conceptual

138) Conceptual

139) Conceptual

140) Conceptual

141) Conceptual

142) Heterozygous individuals (HbA HbS) are clinically diseased under normal oxygen tension.

143) 3 Genotypes, 3 Phenotypes

144) Both (A) and (R) are true and (R) is the correct explanation of (A).

145) Statements I and II are correct; III is incorrect

146) A – C – B

147) Complete dominance for seed shape and incomplete dominance for starch grain size by the same gene.

148) Thomas Hunt Morgan and colleagues

149) Conceptual

150) α-Thalassemia is controlled by two closely linked genes HBA_1 and HBA_2 on chromosome 16.

151) 500 meiotic and 1600 mitotic

152) Central cell

153) Epidermis → Endothecium → Middle layers → Tapetum; Tapetum cells possess dense

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cytoplasm and generally more than one nucleus.

154) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

155) Insects or wind, above water level

156) 5n and 3n

157) Dioecious condition (e.g., Papaya)

158) Proliferation of diploid nucellar cells protruding into embryo sac

159) Both (A) and (R) are true and (R) is the correct explanation of (A).

160) Statements I and II are correct; III is incorrect

161) All seeds lose viability within a few weeks of shedding under natural conditions.

162) Pollen grains are long, ribbon-like, and lack exine/have mucilaginous sheath

163) Guiding the entry of the pollen tube into the synergid

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164) Epiblast

165) Hybrid characters are preserved without segregation, eliminating the need to buy hybrid seeds every year.

166)

The plane of one base pair stacks over the other in double helix. This in addition to H-bonds confers stability of the helical structureThe plane of one base pair stacks over the other in double helix. This in addition to H-bonds confers stability of the helical structure

167) NCERT Pg.# 98

168)

Conceptual

169)

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Conceptual

170) **Question Explanation:** The question asks about the pitch of the DNA helix, i.e., the length of one complete turn of the helix.
Concept : DNA Structure And Features
Solution : Pitch = distance for one complete turn = 10 base pairs Distance between base pairs = 0.34 nm → total pitch = 0.34 × 10 = 3.4 nm

Final answer : Option (1)

171) **Question Explanation:** The question checks correctness of two statements about human DNA content and nitrogenous bases.
Concept : DNA Structure And Features
Solution: Statement-I: Human haploid DNA content = 3.3×10^9 bp. The statement says diploid = 3.3×10^9 bp → Incorrect (diploid is 6.6×10^9 bp).Statement-II: Cytosine is present in both DNA & RNA, and thymine is present in DNA → Correct.

Final answer : Option (3)

172)

Question Explanation :
The question is asking about identifying the scientists who originally put forward the idea or hypothesis that DNA replicates in a semiconservative manner.

Concept :
This question is based on **DNA replication**

Solution :
James Watson and Francis Crick, in their 1953 paper describing the double helix structure of DNA, noted that the specific pairing of bases immediately suggested a copying mechanism. They proposed that each strand of the parent DNA molecule could serve as a template for the synthesis of a new complementary strand, leading to two new DNA molecules, each with one original and one new strand. This is the essence of the semiconservative replication scheme.

Final Answer :
Option (1)

173) Theory based

174)

Theory based

175) NCERT Based

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176) Question Explanation: Discovery and structure facts.
Concept: DNA Structure : History of discovery.
Solution: X-ray diffraction data of Maurice Wilkins and Rosalind Franklin helped Watson and Cricks propose the double helix model.
Final Answer : Option (3)

180)

Conceptual

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